Learning Verified Monitors for Hidden Markov Models

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Abstract. Runtime monitors for hidden Markov models obtain a sequence of observations and must classify whether the system is currently deemed safe. A system is deemed safe, if the probability of being in a dangerous state conditioned on these observations is below some threshold. The key contribution is the first correct-by-construction synthesis method for such monitors, represented as finite state machines. The contribution combines four ingredients: First, we establish the coNP-hardness of checking whether a finite state machine is a correct monitor, i.e., a monitor without misclassifications. Second, we provide a reduction that reformulates the search for misclassifications into a standard probabilistic system synthesis problem. Third, we integrate the verification routine into an active automata learning routine to synthesise correct monitors. Fourth, we provide a prototypical implementation that shows the feasibility and limitations of the approach on a series of benchmarks.

1 Introduction

Runtime assurance is an essential ingredient in the deployment of safe autonomous systems. Runtime monitors provide assurance by flagging potentially dangerous system behavior by observing a system execution. More precisely, a monitor receives a stream of observations about the system and outputs a verdict, e.g., it raises an alarm that the system has left some safety envelope. A monitor is correct if it correctly raises such alarms based on a formal specification. Various challenges in creating correct runtime monitors for (semi-)autonomous systems have been identified [**empty citation**], such as: (1) the state of the system is only partially observable, i.e., the stream of observations comes from sensor readings and does not uniquely identify the state of a system, (2) the behavior of the system and/or the sensors may be subject to probabilistic uncertainty, (3) the monitor itself is subject to resource constraints (time, memory, etc), and (4) the monitor is itself safety-critical and should therefore be subject to extensive validation. Challenges (1,2) can be (partially) addressed by modelling the system as a hidden Markov model (HMM), Challenges (3,4) can be (partially) addressed by representing a monitor as, e.g., a (small) finite state machine. Concretely, this paper focuses on the following question: *Is a given finite state machine a correct monitor for a given HMM?* The paper studies the complexity of this problem, provides a practical verification procedure, and embeds this verification procedure into a learning framework.

Luko: Mention that the system we are monitoring is not a black box. We have full access to the system during learning and verification. Other Runtime verification papers often assume the system is a black box.

Linus: What does partially mean here? Perhaps a footnote that says you discuss this somewhere?

What are HMMs? In this paper, we assume that the system including its sensors is adequately described as a discrete Hidden Markov Model (HMM). Markov chains (MCs) describe system behavior subject to probabilistic uncertainty. Paths through an MC are sequences of states that describe system executions. HMMs extend MCs by labelling their states with colors (aka *observations*). Intuitively, the observations can be used to model the information that the monitor receives. In HMMs, every path can be lifted to a sequence of observations, which we call a *trace*. The trace associated to a system execution describes the information received by the monitor.

Linus: Maybe move after What are correct monitors. I feel like "what are correct monitors" is a question that immediately comes up, while risk of a trace is not.

Linus: How does this relate to the problem statement?

Risk of a trace. In runtime monitoring, an essential task is to estimate whether the current system is dangerous. In the simplest setting, a specification describes whether a set of states is dangerous. The risk of a trace is then the probability of ending in a dangerous state, conditioned on the fact that the system execution matches the trace. For a given trace, we may compute this risk, e.g., either via a reduction to model checking [**empty citation**] or by a (forward) filtering that tracks a distribution over the current states [**empty citation**].

What are correct monitors? Often, we are not interested in this precise risk, but rather want to partition the set of horizon-bounded traces into safe and unsafe traces: Unsafe traces are traces whose risk does exceed some fixed *acceptable threshold*, all other traces are safe. Indeed, a *correct* monitor obtains a trace and raises an alarm iff a trace is not safe. The key verification problem in this paper asks whether a monitor, represented as a deterministic finite automaton, accepts (i.e., raises an alarm) exactly the unsafe traces. We note that in this paper, we only consider traces that are bounded by some fixed horizon.

Illustrative toy example. SJ: TODO make more concrete, add Fig. As an running example, we consider an oversimplified car driving scenario, loosely inspired by runtime monitors obtained from high-fidelity simulations [**empty citation**]. A car can be in three states: It can be on a dry road, on an icy patch, or it can drifted off the road. The HMM in Figure 3 describes the situation, where a car alternates between dry and icy road segments, and where being on a icy (dry) segment positively affects the probability that the next segment is icy (dry). When on an icy road, there is a higher probability to drift off the road. Our sensor can detect dry roads, but cannot distinguish between icy roads and off-road conditions. We can define the trace risk as the probability to be off-road, conditioned on a a trace, such as $dry \cdot icy \cdot icy$. In Fig. 3b, monitor $\mathcal{A}$, we depict a finite automaton that describes the set of traces that end with two consecutive icy patches. The key verification question in this paper is now: Is this a correct monitor for threshold 0.9? Put differently, does it raise an alarm iff the trace risk is above 0.9.

SJ: I parked this figure here from the proposal

SJ: Do we also want a small figure of a system feeding observations into a monitor?

Novel complexity results. Deciding whether a monitor is correct is coNP-complete (Theorems 4). In particular, we study the dual problem that asks whether a monitor misclassifies at least one trace. We consider a simpler variant of this

problem that asks whether there is a bounded trace in the HMM that is unsafe. [1] The problem is clearly in NP (assuming that the bound is unary encoded). We show that this simplified problem is also strongly NP-hard (Lemma 5) and the corresponding optimization problem, that asks to approximate the risk of the trace with the heighest risk, is APX-hard (Lemma 6). Intuitively, APX-hardness means that the existence of an efficient approximation algorithm would have major complexity-theoretic implications.

Our verification approach. While deciding whether a monitor is correct is thus in general not tractable, we suggest to utilise recent advances in the synthesis of probabilistic systems. We concentrate on proving the absence of missed alarms, i.e., we concentrate on showing that the monitor correctly identifies every unsafe trace. A similar reduction works for showing the absence of spurious alarm. First, for a single trace, computing the risk can be reduced to computing a reachability probability in a MC that is intuitively a kind of product between a DFA that accepts exactly the trace and the HMM [**empty citation**]. Thus, it is natural to reduce our problem to the question: Is there a DFA (accepting exactly one trace, accepted by our monitor) such that the probability in the product MC exceeds a threshold. If the answer is no, then the monitor has no-missed-alarms. This decision problem can be solved using (exact) probabilistic model synthesis methods [**empty citation**] as implemented in the tool Paynt [**empty citation**][2]. We discuss our verification approach in Seciton 3.

Linus: The term spurious alarm only appreas here, maybe describe it instead?

Linus: emph?

Our learning approach. Ultimately, we do not only want to verify the correctness of a monitor, but we want to construct such a monitor. Above, we already mentioned that the monitors are formal languages. When considering bounded trace, these monitors are regular, and can thus be captured using DFAs. Inspired by a recent short paper [**empty citation**], we investigate learning these monitors: We choose to do this using active automata learning [5, 28], which closely relates to other popular oracle-guided inductive synthesis loops [**empty citation**]. To

Linus: This was not mentioned yet

[1] The results also apply to checking for safe traces.

[2] Alternatively, in Sec. 3, we show that we can construct a product of an acyclic DFA accepting sets of traces and the HMM. However, this yields partially observable MDP, where we must search for memoryless policies, similar to a construction in [9]. This problem is equivalent to the synthesis problem and can therefore also be solved using Paynt [3].

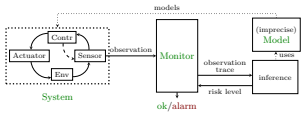


Fig. 1: White-box monitoring of systems

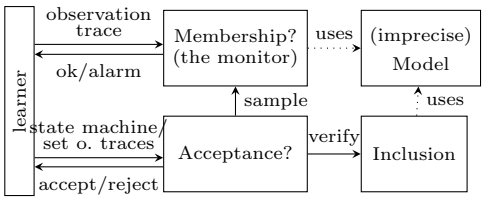
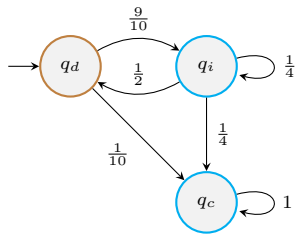
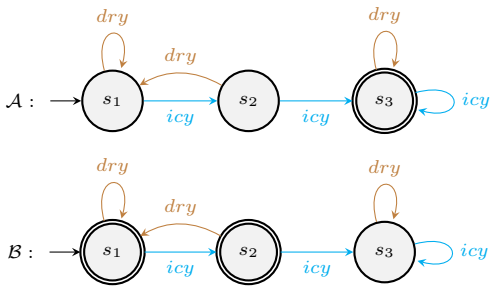


Fig. 2: Learning monitors.



(a) Example HMM, where the cyan states q_i and q_c have the observation *icy* and the brown state q_d has the observation *dry*.



(b) Two example monitors for the HMM in Fig. 3a, correct monitor $\mathcal{A}$ and faulty monitor $\mathcal{B}$. $\mathcal{A}$ raises an alarm as soon as two consecutive steps on an *icy* section have been observed. While $\mathcal{B}$ raises an alarm if less than two consecutive steps on *icy* road has been observed.

Fig. 3: The central example model as a HMM with both a correct monitor and a faulty monitor for a threshold of 0.8.

use active automata learning (AAL), we must provide a *membership oracle* that decides for a single trace whether the risk of the trace exceeds a threshold and an *equivalence oracle* that decides whether a given hypothesis monitor is indeed correct. The membership oracle can be implemented by computing the risk for a single trace as above and comparing it to a threshold, while the equivalence oracle can be implemented by checking for the absence of missed and false alarms. Practically, some minor modifications are necessary to embed our approach in AAL framework: We have to handle the finite horizon, resolve some traces that are not relevant for the monitor, and we combine conformance and equivalence queries to boost performance.

Putting it together. In Fig. **??**, we outline our proposed framework to learn a monitor. SJ: TODO explain framework

Empirical evaluation. SJ: TODO summarise results of the empirical eval here.

Contributions. In summary, this paper provides a first framework to learn runtime monitors — encoded as finite automata — that are verified to be correct with respect to a given HMM. The main contributions are as follows: (1) We study monitor verification and show complexity-theoretic hardness of the underlying problem, even in an approximative setting. (2) We reduce the monitor verification to probabilistic model synthesis problems that exhaustively search for a counterexample. (3) We learn monitors by using the verifier above to answer equivalence queries in conjunction with active automata learning. (4) We demonstrate the feasibility and limitations of both the verifier and the learner on a series of benchmarks. Our prototype can handle HMMs with thousands of states, up to a hundred observations, and monitors with over 100 states.

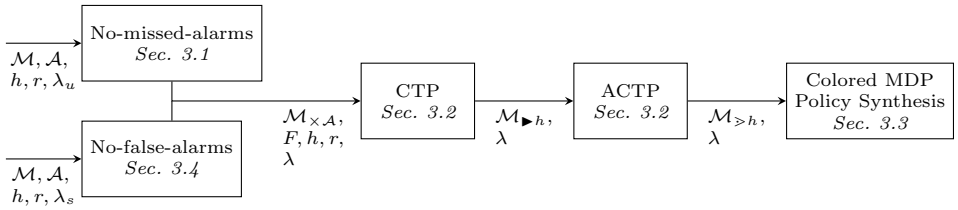


Fig. 4: Overview of proof of Thm. 1, containing the different intermediate problems, together with the inputs for each problem.

2 Formal Problem Statement

Let X be a finite set. A distribution μ over X is a mapping $\mu\colon X \to [0,1]$ such that $\sum_{x\in X} \mu(x) = 1$. The set of all distributions over X is written $\Delta(X)$.

SJ: In the preamble, there was a definition that everything should be displaystyle. I removed it for good reasons, but it may break things.

DFAs. A *deterministic finite automaton* (DFA) is a tuple $\mathcal{A} = (Q, \Sigma, \delta, \iota, F)$. Q is a finite set of *states*, Σ is an *alphabet*, $\delta\colon Q \times \Sigma \rightharpoonup Q$ is a (partial) *transition function*, ι is the *initial state*, and $F \subseteq Q$ is a set of *accept* (or: *final*) states. Let ε denote the empty word. We lift the transition relation to words, which is defined recursively: $\delta^*(q, \varepsilon) = \varepsilon$ if $a \in \Sigma$ and $\delta^*(q, w \cdot a) = \delta(\delta^*(q, w), a)$. The *language* $\mathcal{L}(\mathcal{A})$ of a DFA $\mathcal{A}$ is the set of all words that end in a final state of the $\mathcal{A}$, $\mathcal{L}(\mathcal{A}) = \{w \in \Sigma^* \mid \delta^*(\iota, w) \in F\}$. We say that $\mathcal{A}$ accepts $w \in \mathcal{L}(\mathcal{A})$.

SJ: Shall we just only use final?

2.1 Models

We introduce MDPs and HMMs: The former are integral part of our approach, and HMMs are crucial to the problem statement. Details can be found in [10].

Definition 1 (MDP). *A* Markov decision process *(MDP) is a tuple* $(S, \iota, Act, \mathbf{P})$ *with a countable nonempty set* S *of* states*, the* initial state $\iota \in S$*, the* partial transition function $\mathbf{P}\colon S \times Act \rightharpoonup \Delta(S)$.

We use $Act(s) = \{a \mid \mathbf{P}(s, a) \neq \bot\}$ as the set of *enabled actions*. We assume no deadlocks, i.e., for every $s \in S$, $Act(s) \neq \emptyset$. A path π is a sequence $s_0 a_0 s_1 \ldots \in (S \times Act)^* \times S$, such that $a_i \in Act(s_i)$ and $P(s_i, a_i)(s_{i+1}) > 0$ for every $i \geq 0$. The set of paths in MDP $\mathcal{M}$ is denoted as $\Pi^{\mathcal{M}}$, the set of finite paths are $\Pi^{\mathcal{M}}_{fin}$, the set of paths of at most length h are $\Pi^{\mathcal{M}}_{h}$, and the set of paths of exactly length h are $\Pi^{\mathcal{M}}_{=h}$. We drop $\mathcal{M}$ whenever it is clear from the context.

SJ: Do we?

A *Markov chain* (MC) is an MDP where $|Act(s)| = 1$ for every state $s \in S$. We simplify notation and write MCs as tuple $(S, \iota, \mathbf{P})$ and write $\mathbf{P}(s)$ to refer to the unique distribution $\mathbf{P}(s, a)$. Paths in a MC can be simplified to be sequences of (only) states. The probability measure $Pr^{\mathcal{M}}$ of an MC $\mathcal{M}$ is the unique probability measure following from the canonical σ-algebra associate with $\mathcal{M}$.

SJ: Check consistency of subscript/superscript!!

Luko: Add reachability probability

Definition 2 (HMMs). *A (risk-labelled) HMM is a tuple* $(S, \iota, \mathbf{P}, Z, obs, r)$ *such that* $(S, \iota, \mathbf{P})$ *is a MC,* Z *is a finite set of observations, and* $obs\colon S \to Z$ *is the (deterministic) observation function*[3]*, and* $r\colon S \to \mathbb{R}_{\geq 0}$ *is the risk function.*

Notions such as paths are lifted from MCs. Furthermore, traces τ are sequences of observations, and the set of traces of a HMM, for which there is non-zero probability path, is the language $\mathcal{L}(\mathcal{M})$ of the HMM. The ith observation of a trace is denoted as $\tau^{(i)}$. We define $Pr(\tau \mid \pi) = 1$ if $obs(\pi) = \tau$ and zero otherwise. The probability of a trace τ is $\sum_{\pi \in \Pi_{\mathcal{M}}} Pr(\pi) \cdot Pr(\tau \mid \pi)$. Finally, the conditional probability on a trace $\tau \in \mathcal{L}(\mathcal{M})$ is defined using Bayes' rule $Pr(\pi \mid \tau) := {}^{Pr(\tau \mid \pi) \cdot Pr(\pi)}/_{Pr(\tau)}$.

Example 1. We consider the HMM from Fig. 3a. The probability $Pr(q_c, q_i, q_i \mid dry, icy, icy)$ is ${}^{Pr(dry, icy, icy \mid q_c, q_i, q_i) \cdot Pr(q_c, q_i, q_i)}/_{Pr(dry, icy, icy)}$. $Pr(dry, icy, icy \mid q_c, q_i, q_i)$ is 1, and $Pr(q_c, q_i, q_i)$ is ${}^{9}/_{40}$. The sum of all paths which observe dry, icy, icy is ${}^{11}/_{20}$. Thus, the total probability of q_c, q_i, q_i given dry, icy, icy is ${}^{9}/_{22}$.

2.2 Formal Problem Statement

Definition 3 (Monitor). *A DFA* $\mathcal{A}$ *is a monitor for HMM* $\mathcal{M}$ *if the alphabet for* $\mathcal{A}$ *coincides with the observations in* $\mathcal{M}$*.*

SJ: $\pi_{\downarrow}$ is not yet defined!

Monitors should accept unsafe traces, which we define via their risk [20]:

Definition 4 (Trace risk, safe/unsafe traces). *Given HMM* $\mathcal{M}$*, the* risk *of* $\tau \in \mathcal{L}(\mathcal{M})$ *is:*

$$R(\tau) := \sum_{\pi \in \Pi^h_{\mathcal{M}}} Pr(\pi \mid \tau) \cdot r(\pi_{\downarrow}).$$

Let $\lambda_s \leq \lambda_u \in \mathbb{R}$ *be the* safe threshold *and an* unsafe threshold*. A trace* $\tau \in \mathcal{L}(\mathcal{M})$ *with* $R(\tau) > \lambda_u$ *is* unsafe*, while* τ *is* safe *if* $R(\tau) < \lambda_s$*.*

We deliberately do not require $\lambda_s = \lambda_u$. By picking $\lambda_s < \lambda_u$, some traces are neither safe nor unsafe. We write $\mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s}$ (and $\mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u}$) for the set of safe (and unsafe) traces of length at most h.

Example 2. We consider the HMM from Fig. 3a, with the risk function assigning 1 to q_c and 0 to all other states. Taking the trace dry, icy, icy, there are three paths which could generate this trace. Two paths end in q_c, one ends in q_i. The paths ending in q_c have a conditional probability of ${}^{13}/_{22}$. Since only these paths have a non-zero risk, the risk of the trace is ${}^{13}/_{22} \cdot r(q_c) = {}^{13}/_{22}$.

Definition 5 (Missed/False alarms). *Given a monitor* $\mathcal{A}$ *for HMM* $\mathcal{M}$*, a horizon* h*, and thresholds* $\lambda_s \leq \lambda_u \in \mathbb{R}$*, the set of* missed alarms *is* $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) := \mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u} \setminus \mathcal{L}(\mathcal{A})$*. The set of* false alarms *is* $\mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) := \mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s} \cap \mathcal{L}(\mathcal{A})$*.*

[3] We use determinstic observation functions for concise definitions. Stochastic observation functions can be expressed via a blowup of the HMM, see e.g. [20].

Definition 6 (Correct monitor). *Given thresholds λ_s, λ_u and horizon h, a monitor $\mathcal{A}$ for $\mathcal{M}$ is* correct *if* $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) = \emptyset = \mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_u}(\mathcal{A})$.

A correct monitor raises an alarm for all unsafe traces and for no safe trace, i.e., missed alarms are false negatives, while false alarms are false positives.

Corollary 1. *A monitor $\mathcal{A}$ is correct iff* $\mathbb{U}^{\leq h}_{\mathcal{M},\lambda_u} \subseteq \mathcal{L}(\mathcal{A}) \subseteq \Sigma^* \setminus \mathbb{S}^{\leq h}_{\mathcal{M},\lambda_s}$.

Problem statements. Given HMM $\mathcal{M}$, thresholds λ_s, λ_u and horizon h:
1. Given monitor $\mathcal{A}$ for $\mathcal{M}$, are there missed alarms, i.e., is $\mathsf{mA}^{\leq h}_{\mathcal{M},\lambda_s}(\mathcal{A}) = \emptyset$?
2. Given monitor $\mathcal{A}$ for $\mathcal{M}$, are there false alarms, i.e., is $\mathsf{fA}^{\leq h}_{\mathcal{M},\lambda_u}(\mathcal{A}) = \emptyset$?
3. Find a correct monitor $\mathcal{A}$ for $\mathcal{M}$ w.r.t. λ_s, λ_u and h.

Problems 1 and 2 together allow checking whether a monitor is correct. Furthermore, a correct monitor must exist, as $\mathbb{U}^{\leq h}$ is finite and thus regular.

Example 3. We discuss the example from Fig. 3 using the correct monitor, $\mathcal{A}$. Given the risk function assigning 1 to q_c and 0 to all other states, the traces $\tau_1 = dry, icy, icy$ and $\tau_2 = dry, icy$ have risks $13/22, 1/10$ respectively. If λ_s is $1/4$ and the horizon h is 3, τ_2 is the trace with maximum risk not accepted by the monitor. Given that its risk is below λ_s, the monitor does not have any missed alarms. Similarly, monitor $\mathcal{A}$ does not have any false alarms for $\lambda_u = 1/4$. Thus, $\mathcal{A}$ is a correct monitor for $\mathcal{M}$ with h, λ_s, and λ_u.

3 Monitor Verification

We present our approach to the monitor correctness problem, which reduces checking the existence of missed alarms to policy synthesis on colored MDPs, defined below. The latter problem is well studied. We first formalize the policy synthesis problem and then discuss the transformation in multiple steps. Here, we focus on showing that there are no missed alarms of exactly the length of the horizon (an adaption of Problem 1). At the end of the section, we generalize our construction to finding false alarms and to traces of length *at most* the horizon.

3.1 Relating Missed Alarms with Color-Consistent Policies

Policy synthesis for an MDP for a property ϕ thus entails finding a policy for an MDP such that the induced MC entails ϕ. A policy for MDP $\mathcal{M}$ is a function $\sigma\colon \Pi^{\mathcal{M}}_{fin} \to Act$, which selects actions for every finite path. An MDP and a policy induce an MC.

SJ: Moved these sentences down, need to integrate them?

Colored (Labelled) MDPs are an extension to MDPs that allow expressing dependencies between states that policies must adhere to. The following definition simplifies the notion of colored MDPs from [**empty citation**], and suffices for our needs:

Definition 7 (Colored MDP). *Given an MDP $\mathcal{M}$ with states S, a* colored MDP *is a tuple $\mathcal{M}^C = (\mathcal{M}, C, c)$, where C is a set of* colors*, and $c\colon S \to C$ maps states to colors.*

Definition 8 (Color consistent). *A memoryless scheduler σ for a colored MDP $\mathcal{M}^C$ is* color consistent[4] *if for states s, s', $c(s) = c(s')$ implies $\sigma(s) = \sigma(s')$.*

The set of all color-consistent schedulers is denoted Σ_c. Policy synthesis for colored MDPs asks to find a color-consistent scheduler such that the reachability probability to a set of target states is above a certain threshold. Policy synthesis for colored MDPs is NP-hard[**empty citation**].

Theorem 1 (Main theorem). *Given a HMM $\mathcal{M}$ with a monitor $\mathcal{A}$, threshold λ_s, horizon h, and risk r, there is a colored MDP $\mathcal{M}^C$ with target states T s.t.*

$$\exists \sigma \in \Sigma_c.\ Pr_{\sigma}^{\mathcal{M}^C}(\Diamond T) \geq \lambda_s \quad \textit{iff} \quad \exists \tau \in \mathcal{L}(\mathcal{M}).\ \tau \in \mathsf{mA}^{=h}_{\mathcal{M},\lambda_s}(\mathcal{A}).$$

Our proof, outlined in this section, is constructive and we show that we can use the construction to find a $\tau \in \mathsf{mA}(\mathcal{A})$, whenever such a τ exists.

Outline of the proof. The proof is a direct consequence of Lemmas 1 to 3 below. Before we discuss these lemmas, we observe that on the left-hand side of Theorems 1, the monitor, horizon, observations, and risk do not occur, they must be encoded into the colored MDP. Furthermore, while missed alarms are defined using conditional probabilities, the policy synthesis problem is over reachability probabilities. We describe our transformation in several steps. In Section 3.2 we encode the monitor into the HMM and transform the HMM to both include the horizon and the risk. In Section 3.3 we resolve the conditioning on traces, while replacing observations from the HMM.

3.2 The (Acyclic) Conditional Trace Probability Problem

First, we show how asking for a missed alarm can be rephrased into the conceptually simpler *conditional trace probability* (CTP) problem. We will further simplify the problem by incorporating both the horizon and risk such that we are left with a problem purely on acyclic HMMs.

CTP problem We take (a mild variation of) a standard product construction [10] between HMM and the DFA. We define as *alarm states* F those states which correspond to not accepting states in the DFA. This is equivalent to taking a product with the complement of the monitor.

Example 4. Figure 5 shows the product of the HMM and monitor $\mathcal{B}$ from Figure 3. Starting in the initial states d and 1, respectively, the HMM can transition to

[4] Colored MDPs with color-consistent policies coincide with memoryless policies for partially observable MDPs. However, POMDPs often consider history-dependent (belief-based) policies. We use *colored MDPs* to avoid any confusion.

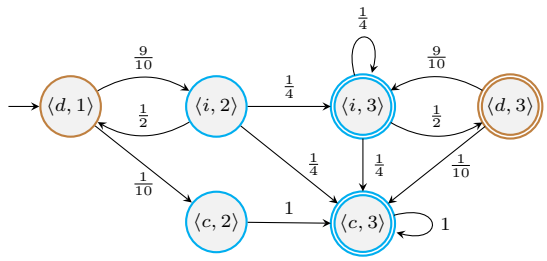


Fig. 5: The central example HMM and faulty monitor $\mathcal{B}$ transformed into the CTP problem. States are named with their HMM state, $\{d, i, c\}$ and their monitor state, $\{1, 2, 3\}$. States in the alarm states set F have been marked accepting.

the i state with probability $9/10$. This is an *icy* state, and thus the monitor takes the *icy* transition to the 2 state. This results in the new product state of i and 2, with a transition from $\langle d, 1\rangle$ to $\langle i, 2\rangle$ with probability $9/10$. The alarm states are any product states $\langle _, 3\rangle$.

Definition 9 (HMM product). *Given a HMM* $\mathcal{M} = (S, \iota^{\mathcal{M}}, \mathbf{P}, Z, obs, r)$ *and monitor* $\mathcal{A} = (Q, \Sigma, \delta, \iota^{\mathcal{A}}, F')$*, the* product HMM $\langle \mathcal{M}_{\times\mathcal{A}}, F\rangle$ *is the HMM* $\mathcal{M}_{\times\mathcal{A}} = (S \times Q, \langle \iota^{\mathcal{M}}, \iota^{\mathcal{A}}\rangle, \mathbf{P}', obs', r')$ *with* $obs'(\langle s, q\rangle) = obs(s)$*,* $r'(\langle s, q\rangle) = r(s)$

$$\mathbf{P}'(\langle s, q\rangle)(\langle s', q'\rangle) = \begin{cases} \mathbf{P}(s, s') & \text{if } \delta(q, obs(s')) = q' \\ 0 & \text{otherwise} \end{cases}$$

And the alarm states $F = S \times F'$*.*

In the product, we can find a trace whose conditional trace probability exceeds a threshold iff there is a missing alarm. We state the decision problem that needs to be solved: It is key to our computational complexity analysis in Seciton 5.

Definition 10 (CTP Decision Problem). *Given HMM* $\mathcal{M}$ *with states* S *and risk* r*, (unary encoded) horizon* h*, alarm states* $F \subseteq S$*, and threshold* $\lambda_s \in \mathbb{R}^+$*,*

$$\exists \tau \in \mathcal{L}(\mathcal{M}).\ \sum\nolimits_{\pi \in \Pi^{\mathcal{M}}_{=h} \mid \pi_{\downarrow} \in F} Pr^{\mathcal{M}}(\pi \mid \tau) \cdot r(\pi_{\downarrow}) \geq \lambda_s.$$

We denote the set of witnesses τ to a CTP decision problem as $\mathsf{CTP}(\mathcal{M}, h, F, \lambda_s)$. The following lemma states the correctness of the transformation and follows directly from the definition of missing alarms and the product with the complement.

SJ: Nevertheless, we should have a (slightly) more formal proof

Lemma 1. *Using the notation from Theorems 1, Definition 9 and Definition 10:*

$$\exists \tau \in \mathsf{CTP}(\mathcal{M}_{\times\overline{\mathcal{A}}}, h, F, \lambda_s) \quad \textit{iff} \quad \exists \tau \in \mathsf{mA}^{=h}_{\mathcal{M},\lambda_s}(\mathcal{A}).$$

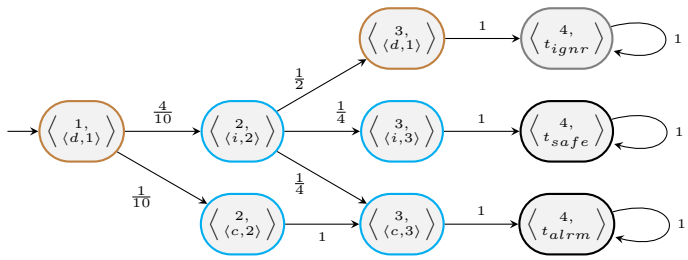


Fig. 6: ACTP, brown and cyan are dry and icy observations. Black is the new z_end observation, and gray is the new z_ignore observation. All states are named with the step i, model state s and monitor state j as $\left\langle \begin{smallmatrix} i, \\ \langle s,j \rangle \end{smallmatrix} \right\rangle$.

ACTP problem We further simplify the problem by unrolling the model along the horizon. We also eliminate the risk function.

Example 5. Our unrolling applied to the HMM from Fig. 5 can be seen in Fig. 6. We unroll the product for horizon three. Thus, the initial state becomes the tuple of step 1 and the initial state from the CTP HMM, $\langle d, 1\rangle$. This state $\langle 1, \langle d, 1\rangle\rangle$ transitions to, e.g., $\langle 2, \langle c, 2\rangle\rangle$. Consider state $\langle 3, \langle i, 3\rangle\rangle$, it is at the horizon and its corresponding CTP state is in the alarm states F, the monitor raises an alarm. The normalized risk for this state is 0, since $r(\langle d, 1\rangle) = 0$. Thus, $\langle 3, \langle i, 3\rangle\rangle$ transitions with probability 1 to the safe state, $\langle 4, t_{safe}\rangle$, and probability 0 to the alarm state, $\langle 4, t_{alrm}\rangle$. The corresponding CTP state of $\langle 3, \langle d, 1\rangle\rangle$ is not in F. It is a horizon state, and thus it transitions to the ignore state, $\langle 4, t_{ignr}\rangle$.

Definition 11 (Unrolling with risk). *The (risk-)unrolled HMM $\mathcal{M}_{\blacktriangleright h}$ of a HMM $\mathcal{M} = (S, \iota, \mathbf{P}, Z, obs, r)$ with horizon h and alarm states $F \subseteq S$ is the HMM $\mathcal{M}_{\blacktriangleright h} = (\{1, \ldots, h\} \times S \cup \{\langle h+1, t_{alrm}\rangle, \langle h+1, t_{safe}\rangle, \langle h+1, t_{ignr}\rangle\}, \langle 1, \iota\rangle, \mathbf{P}', Z \cup \{z_{end}, z_{ignore}\}, obs')$, with the new transition and observation function:*

$$\forall_{i \in \{1,\ldots,h-1\}} \quad \mathbf{P}'(\langle i, s\rangle)(\langle i+1, s'\rangle) = \mathbf{P}(s)(s')$$

$$\mathbf{P}'(\langle h, s\rangle)\,(\langle h+1, t\rangle) = \begin{cases} \frac{r(s)}{\max_{s \in S} r(s)} & \text{if } s \in F \wedge t = t_{alrm} \\ 1 - \frac{r(s)}{\max_{s \in S} r(s)} & \text{if } s \in F \wedge t = t_{safe} \\ 1 & \text{if } s \notin F \wedge t = t_{ignr} \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbf{P}'\,(\langle h+1, t\rangle)\,(\langle h+1, t\rangle) = 1$$

$$obs'(\langle i, s\rangle) = obs(s)$$

$$obs'(\langle h+1, t\rangle) = \begin{cases} z_{end} & \text{if } t \in \{t_{alrm}, t_{safe}\} \\ z_{ignore} & \text{if } t = t_{ignr} \end{cases}$$

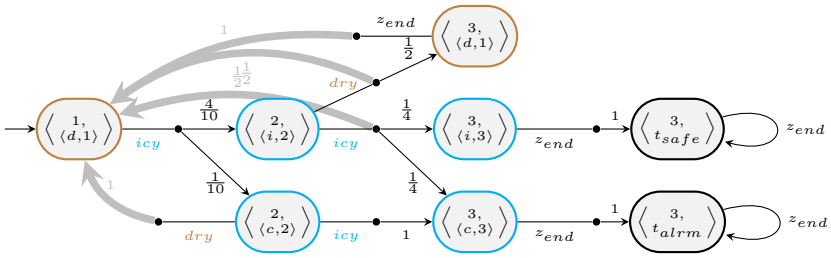


Fig. 7: The central example transformed into an mdp to be used as the model in the MDP policy synthesis problem. For conciseness, all unreachable states are removed. Also any actions which return for all states in the same step are removed.

For the first h steps, the unrolling is standard. At the horizon we transition to the three dedicated states t_{alrm}, t_{safe} and t_{ignr}[5] according to the risk.

Lemma 2. *Given an HMM $\mathcal{M}$, horizon h, alarm states F, and threshold λ_s, $\exists\lambda \in (0,1]$:*

$$\exists\tau \in \mathsf{CTP}(\mathcal{M}, h, F, \lambda_s) \quad \textit{iff} \quad \exists\tau \in \mathcal{L}(\mathcal{M}_{\blacktriangleright h}).\ \sum_{\pi\in\Pi^{\mathcal{M}}} Pr^{\mathcal{M}}(\pi\cdot t_{alrm} \mid \tau\cdot z_{\mathsf{end}}) \geq \lambda,$$

where z_{end} is the final observation and t_{alrm} is the alarm state.

SJ: I wonder whether we should simplify notation on the right hand side. I am not too fond of the z and s_a symbols.

SJ: Put a short proof sketch in the appendix

3.3 Reduction to Consistent Policy Synthesis

For Lemma 2, we must find a trace such that a conditional reachability probability exceeds a threshold. We reformulate this into a policy synthesis problem in MDPs. The transformation combines two ideas: First, in every state, the policy can select the next observation, inspired by [**empty citation**]. Second, we reformulate conditional reachability probabilities by (standard) regular reachability probabilities, using a construction in [**empty citation**].

Example 6. We transform the HMM from Fig. 6 to the Colored MDP from Fig. 7. Consider $\langle 2, \langle i, 2\rangle\rangle$ in original the HMM. The next observation is either *dry* or *icy*, which corresponds to two new actions for $\langle 2, \langle i, 2\rangle\rangle$ in the colored MDP. The *dry* action transitions to the ensuing states with the *dry* observation. The remaining probability, $1/2$, for the *dry* action is directed to the initial state. This is similarly applied to the *icy* action. State $\langle 2, \langle c, 2\rangle\rangle$ does not reach any *dry* states, we still add a *dry* action redirecting to the initial state, since we want all states with the same step value to have the same actions. State $\langle 3, \langle d, 1\rangle\rangle$ in the HMM only has

[5] We add an $\langle h+1, t_{ignr}\rangle$ state, instead of redirecting all traces we don't care about to the $\langle h+1, t_{safe}\rangle$ state, to more easily modify the transformation for the no-false-alarms problem.

a transition to an z_{ignore} state. We fully remove the z_{ignore} state and the z_{ignore} observation in the MDP. Finally, we make sure all states with step 3 have the same actions, and thus add the z_{end} action redirecting to the initial state.

Definition 12. *Given an acyclic HMM $\mathcal{M}_{\blacktriangleright h} = (S, \iota, \mathbf{P}, Z, obs, r)$ as in Definition 11, we define the colored MDP $\mathcal{M}_{\gtrdot h} = ((S', \iota, Act, \mathbf{P}'), C, c)$ with $Act = Z \setminus \{z_{ignore}\}$,*

$$\mathbf{P}'(\langle i, s\rangle, z)(\langle j, q\rangle) := \begin{cases} \mathbf{P}'(\langle i, s\rangle)(\langle j, q\rangle) & \text{if } obs(\langle j, q\rangle) = z, \\ \sum_{(\langle k,q'\rangle)\in S' | obs(\langle k,q'\rangle)\neq z} \mathbf{P}'(\langle i, s\rangle)(\langle k, q'\rangle) & \text{if } (\langle j, q\rangle) = \iota, \\ 0 & \text{otherwise} \end{cases}$$

$C = \{1, \ldots, h\}$ and c is given such that $c(\langle i, s\rangle) = i$.

Thus, we transition normally to a state if the action and observation of the target state are equal, otherwise we set the transition probability to zero. All the remaining probability mass is redirected towards the initial state.

Remark 1. All states gain all actions in $\mathbf{P}''$, we can later prune any actions for all states S^C of a color C when these actions only redirect to the initial state from S^C.

The above construction allows for conditioning on a trace τ by constructing a policy σ that selects the ith observation of τ in the state with step i.

Definition 13 (Trace consistent policy). *Given an MDP $\mathcal{M}_{\gtrdot h}$ as defined in Def. 12. Together with a trace $\tau \in Z^\star$. A* trace consistent policy *is of the form*

$$\sigma_\tau(\langle i, s\rangle) = \begin{cases} \tau^{(i)} & \text{if } i \leq |\tau|, \\ end & \text{otherwise.} \end{cases}$$

Using the coloring as described in Def. 12, the trace consistent policies coincide with the color consistent policies. Finding a missed-alarm traces now reduces to solving the following color consistent policy synthesis problem:

$$\exists \sigma \in \Sigma_c . Pr^\sigma(\Diamond \langle h + 1, t_{alrm}\rangle) > \lambda_s$$

Which is solved using PAYNT [4].

Lemma 3. *Given an HMM $\mathcal{M}$, horizon h, alarm states F and threshold λ:*

$$\exists \sigma \in \Sigma_c . Pr_\sigma^{\mathcal{M}_{\gtrdot h}}(\Diamond T) \geq \lambda \quad \text{iff} \quad \exists \tau \in \mathcal{L}(\mathcal{M}_{\blacktriangleright h}). \sum_{\pi \in \Pi^{\mathcal{M}_{\blacktriangleright h}}} Pr^{\mathcal{M}_{\blacktriangleright h}}(\pi \cdot t_{alrm} \mid \tau \cdot z_{end}) \geq \lambda,$$

Luko: Fix hbox

Corollary 2. *There exists a map $t(\sigma_\tau) = \tau$ which, given a trace consistent policy σ_τ, finds its associated trace τ.*

3.4 Adapting to No-False-Alarms and Smaller Traces

Luko: Reference Prob. 2 somewhere here.

- Add end transitions to non horizon states.
- We want to minimize alarm probability, not possible.
- Instead maximize safe probability.

Traces of Length up to the Horizon The aproach for Thm. 1 works only for traces of exactly the length of the horizon.

Luko: Something about it being hard for ACTP but easy for Pol Syn

We transform Def. 12 such that the following theorem holds:

Theorem 2. *Luko: Something relating the modified transformation to missing alarms on $\leq h$*

Proof. We modify the transformation from Thm. 1 in the policy synthesis step. We add a new initial state, ι', it gets a seperate coloring from all other states. This state contains an action for each possible length l of a trace up to the horizon. The actions now transition to the state $\langle l, \iota \rangle$.

$$\mathbf{P}'(\iota', l)(\langle l, \iota \rangle) = 1$$

We now show this transformation is correct.

In order to verify that there are no-missed-alarms for all $l < h$, we could use the transformation from Thm. 1 with the horizon equal to all $l < h$. This would neccesitate doing policy synthesis for h colored MDPs.

$$\exists_\tau \exists_{l \leq h} Pr^{\sigma_\tau}[\Diamond alarm] > \lambda_s$$

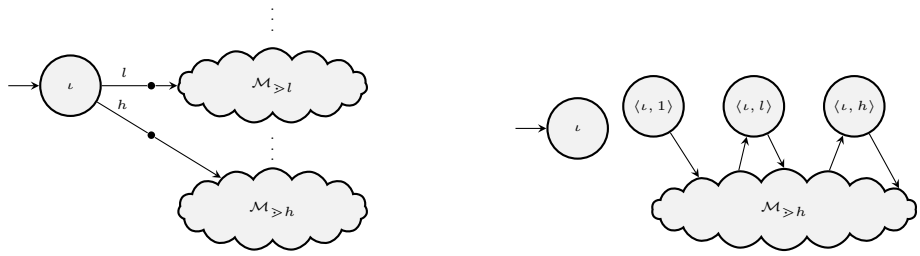
If any of these h policy synthesis problems can find a policy σ_τ, there exists a trace $\tau \in \mathsf{mA}_{\leq h}$.

We combine these h colored MDPs into one colored MDP in the following way. We add a new initial state, and give it h actions $\{1, \ldots, h\}$. Action $l \in \{1, \ldots, h\}$ points, with probability 1, to the initial state of colored MDP $\mathcal{M}_{\geqslant l}$. This is equivalent to solving policy synthesis on the h individual MDPs.

We now note that for any $l < l' \leq h$ the initial state $\langle 1, \iota^{CTP} \rangle$ of $\mathcal{M}_{\geqslant l}$ is bisimilar to the state $\langle l' - l + 1, \iota^C TP \rangle$ in $\mathcal{M}_{\geqslant l'}$. We now claim that the result of bisimulation minimization on this combined colored MDP is described by the transformation described at the start of the proof.

Luko: Maybe note that the $\mathcal{M}_{\geqslant l}$ only differed when they where in the CTP stage by the horizon, nothing else.

Example 7. The transformation in Thm. 2 can be seen in Fig. 8



(a) (b)

Fig. 8

No-False-Alarms Problem We modify the transformation from Thm. 1 such that it solves the no-false-alarms problem. The no-false-alarms problem differs in two ways from the no-missed-alarms problem. Fristly, instead of finding a trace not in the monitor, we have to find a trace in the monitor. Secondly, this trace should have a conditional probability below the threshold. We solve these one at a time after stating Thm. 1 for no-false-alarms.

Theorem 3. *Given a HMM $\mathcal{M}$ with a monitor $\mathcal{A}$, threshold λ_u, horizon h, and risk r, there exists a colored MDP $\mathcal{M}^C$ and a λ such that*

$$\exists \sigma \in \Sigma_c . Pr_{\sigma}^{\mathcal{M}^C}(\Diamond T) \leq \lambda \quad \textit{iff} \quad \exists \tau \in \mathsf{fA}^{=h}_{\mathcal{M},\lambda_s}(\mathcal{A}).$$

Proof. We reuse the transformation from Thm. 1 with the following modifications. We no longer take the complement of the monitor. Instead of doing reachability on the alarm state, we instead do reachability on the safe state.

4 Learning Correct Monitors

Luko: Maybe refrence Prob. 3 somewhere here.

We now describe how to *learn* correct monitors, by combining the automata learning framework based on a Minimally Adequate Teacher (MAT) [5] and the verification approach above.

MAT framework. We briefly recap the MAT framework, for details see [28]. A minimally adequate teacher answers two types of questions: A membership query (MQ), which in our setup means *should a trace be accepted by the monitor?*, and an equivalence query (EQ), *is this monitor correct?* Furthermore, if the answer to an equivalence query is negative, we must provide a counterexample that witnesses why the monitor is not correct. Various algorithms implementing the MAT framework for DFA learning exist. For the purpose of this paper, we use the L* algorithm [5] to learn a monitor. The learner asks MQs to the teacher until a hypothesis monitor can be constructed which is consistent with the MQs. Once such a hypothesis is constructed, the monitor is verified using an EQ.

Verification as a MAT. To learn a monitor $\mathcal{A}$, we provide the HMM $\mathcal{M}$, a risk function r, a horizon h, a learning threshold λ_l, and the safe and unsafe thresholds λ_s and λ_u. The additional risk threshold λ_l is to define an MQ whenever $\lambda_s \neq \lambda_u$. For the MQ, each trace must be unambigiously be defined as safe or unsafe: the MAT framework does not allow for flagging certain traces as *don't care*. However, if a trace has a risk between λ_s and λ_u, the trace is neither safe nor unsafe. Likewise, the MQ must also be defined for traces $\tau \notin \mathcal{L}(\mathcal{M})$ or traces longer than the horizon. We thus adapt the notion of safe traces from Def. 4.

Definition 14. *Given any trace $\tau \in Z^\star$, and a horizon h, membership query MQ_{λ_l} is a function such that $MQ_{\lambda_l}(\tau)$ is unsafe iff $\tau \in \mathcal{L}(\mathcal{M})$[6] and $\tau \in \mathbb{U}_{\lambda_l}^{\leq h}$.*

Such a function for $\mathrm{MQ}_{-\lambda_l}$ can be defined by keeping track of the probability of being in each state after every observation from the trace [20].

We use the verification described in Sec. 3 to define the EQ.

Definition 15. *Given a HMM $\mathcal{M}$, and a monitor $\mathcal{A}$, an EQ_{λ_s,λ_u} is a function g such that $g^{\mathcal{M}}(\mathcal{A})$ holds if $\mathcal{A}$ is correct for $\mathcal{M}$ with λ_s, and λ_u (in the sense of Def. 6). And returns a counterexample trace if $\mathcal{A}$ is not correct.*

The EQ requires checking both for no-missing-alarms, and for no-false-alarms. Each check follows the steps as described in Sec. 3.

Lemma 4. *Given a MAT with an EQ_{λ_s,λ_u} and an MQ_{λ_l}, a monitor learned with $L^\star$ is correct as long as $\lambda_s \leq \lambda_l \leq \lambda_u$.*

Proof. If, while learning a monitor $\mathcal{A}$ a trace τ is deemed *safe* by MQ_{λ_l} it cannot be given as a counterexample by $\mathrm{EQ}_{\lambda_s,\lambda_u}$ on $\mathcal{A}$, since $\lambda_s \leq \lambda_l \leq \lambda_u$, and $\tau \notin \mathcal{A}$ by $\mathrm{L}^\star$. Similarly, MQ_{λ_l} and $\mathrm{EQ}_{\lambda_s,\lambda_u}$ also agree on *unsafe* traces. Now, by correctness of $\mathrm{L}^\star$, a learned monitor $\mathcal{A}$ has to be correct according to $\mathrm{EQ}_{\lambda_s,\lambda_u}$, and thus correct in the sense of Def. 6.

When $\lambda_s < \lambda_u$, the EQ has an inconclusive area given by the interval (λ_s, λ_u). This means that our EQ strictly speaking does not check for equivalence, but simply accepts any correct monitor. We investigate the effect of this inconclusive area in Sec. 6(**Q4**).

Conformance queries. An alternative to the EQ in Def. 15 is a conformance query [15]. It tests a monitor by sampling traces from the HMM and checking if the MQ and the monitor agree. If all the monitor and the MQ don't agree on a trace, it is given as a counter example. In our approach we use a hybrid of the two EQs. Monitors produced early in the learning process often contain many missed alarms and false alarms. Verification can find them, however, applying the transformation from Sec. 3 has a constant cost. Conformance queries can often find a counter example faster if one has a high probability of occurring.

[6] Defining traces $\tau \notin \mathcal{L}(\mathcal{M})$ safe is an arbitrary decision.

SJ: Forward filtering in HMMs should have a different citation as well.

SJ: Not urgent: Couldnt this be done with the techniques in section 3 as well (although inefficiently)?

Luko: Yes, you can create a policy from a trace and evaluate the induced MC.

SJ: give the return type (true or trace)

SJ: Be more specific, it is either a missed alarm or a false alarm?

Loes: Why not let $\mathrm{EQ}_{\lambda_s,\lambda_u}$ be the function instead of g?

SJ: agree

SJ: no need to repeat these steps, so i commented out the next line

Luko: This is a bit short through the corner. But don't know how detailed to be.

Luko: We no longer have the questions

5 Computational Complexity

Theorem 4. Is a monitor correct? *(w. unary coded horizon) is coNP-complete.*

In fact, we study the dual to this problem, i.e., checking the existence of a counterexample. We call this problem *monitor co-verification*. For monitor co-verification, *membership* in NP follows as false alarms or a missed alarm (of length up to horizon, i.e., polynomially sized) are witnesses. Verifying whether a trace is a false or missed alarm can be done in polynomial time, by checking whether the automaton accepts it and computing the trace risk (see Sec. 4).

To establish NP-hardness, we consider the CTP problem from Definition 10. As a solution to the monitor co-verification problem solves the CTP problem (using a trivial monitor), this implies NP-hardness of the former problem.

Lemma 5. *The CTP Decision Problem is NP-hard.*

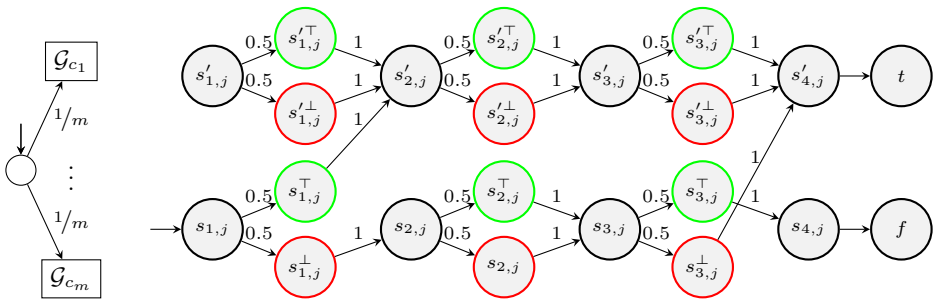
The lemma holds even for $\lambda = 1$ and acyclic HMMs. We reduce the problem from 3CNF. We outline the construction below.

Consider a 3CNF formula $\varphi = \bigwedge c_1 \dots c_m$ over variables X, $|X| = n$, with clause $c_j = \ell_j^1 \vee \ell_j^2 \vee \ell_j^3$ and each literal $\ell_j^i \in \{x, \neg x \mid x \in X\}$. We transform this into a CTP instance with $\lambda = 1$ and an acyclic HMM $\mathcal{M}_\varphi$ with observations $Z = \{\#, \bot, \top\}$. The only state with positive risk is a dedicated state t with $r(t) = 1$. The crux of the construction is that there is a trace with risk 1 iff there is a satisfying assignment to φ. In the constructed HMM, there is a trace with risk 1 iff there is a trace where all corresponding paths end in state t.

SJ: make sure that corresponding paths are the right formal notion

Before we give a formal definition of $\mathcal{M}_\varphi$, we give some intuition. We represent assignments $\alpha \colon X \to \{\bot, \top\}$ by traces through the HMM $\#\#\cdot\alpha(x_0)\cdot\#\cdot\alpha(x_1)\cdot\dots\cdot\#$. We create gadgets for every clause. The gadgets are connected as in Figure 9a: That is, intuitively, $\mathcal{M}_\varphi$ randomly selects a clause c_j with probability $1/m$ and transitions into gadget $\mathcal{G}_j$ defined below. Next, we show that the gadget reaches positive risk in every gadget only if the corresponding clause is satisfied by the assignment.

The gadget $\mathcal{G}_j$ intuitively 'evaluates' c_j with respect to an assignment α, by starting in $s_{1,j}$ with a gadget for this clause, as exemplified in Fig. 9b. We enter the gadget with trace $\# \cdot \alpha(x_0) \cdot \# \cdot \alpha(x_1) \cdot \dots \cdot \#$, i.e., the first observation has been matched by the initial state. A path (or its trace) through $\mathcal{G}_j$ 'reads' variable x_i in state $s_{i,j}$ and transitions to $s_{i,j}^\top$ or to $s_{i,j}^\bot$. However, only one of these paths corresponds to the trace. Thus, in every gadget, there is only one path that corresponds to a given trace and so they can be used interchangeably. State $s_{i,j}$, $s_{i,j}^\top$ and $s_{i,j}^\bot$ have observations $\#, \top, \bot$, respectively. We note that a path reaches $s_{i,j}^\top$ if $\alpha(x_i) = \top$ and (analogously for $\bot$). From there onwards, if the literal x_i (or $\neg x_i$) occurs in clause c_j, we transition from $s_{i,j}^\top$ (or $s_{i,j}^\bot$, resp.) to $s'_{i+1,j}$, otherwise, we transition to $s_{i+1,j}$. A path ends in t only via a state $s'_{i,j}$, which is only possible if the path visits a state corresponding to a literal in the clause. Thus, (conditionally) reaching state $s'_{i,j}$ with positive conditional probability means that the partial assignment $\alpha(x_1), \dots, \alpha(x_i)$ satisfies clause c_j,



(a) HMM. (b) Gadget demonstrated for $x_1 \vee \neg x_3$

Fig. 9: Illustrations for Lemma 5, with m clauses and variables x_1, x_2, x_3.

while reaching $s_{i,j}$ with positive conditional probability means that the clause is either unresolved or unsatisfied given the partial assignment.

Formally, we construct the HMM $\mathcal{M} = (S, \iota, \mathbf{P}, Z, obs, r)$ and set $\lambda = 1$:

- $S = \{s_{i,j}, s^{\bot}_{i,j}, s^{\top}_{i,j}, s'_{i,j}, s'^{\bot}_{i,j}, s'^{\top}_{i,j} \mid i \in \{1, \dots, n+1\}, j \in \{1, \dots, m\}\} \cup \{s_\iota\}$.
- $\mathbf{P}$ is given such that for all $i \in \{1, \dots, n\}, j \in \{1, \dots, m\}$:
  - $\mathbf{P}(s_\iota, s_{1,j}) = \frac{1}{m}$,
  - $\mathbf{P}(s_{i,j}, s^{\top}_{i,j}) = \mathbf{P}(s_{i,j}, s^{\bot}_{i,j}) = \frac{1}{2} = \mathbf{P}(s'_{i,j}, s'^{\top}_{i,j}) = \mathbf{P}(s'_{i,j}, s'^{\bot}_{i,j})$,
  - $\mathbf{P}(s^{\top}_{i,j}, s'_{i+1,j}) = [x_i = \ell^k_j$ for some k$]$, $\mathbf{P}(s^{\top}_{i,j}, s_{i+1,j}) = [x_i \neq \ell^k_j$ for all k$]$,
  - $\mathbf{P}(s^{\bot}_{i,j}, s'_{i+1,j}) = [\neg x_i = \ell^k_j$ for some k$]$, $\mathbf{P}(s^{\bot}_{i,j}, s_{i+1,j}) = [\neg x_i \neq \ell^k_j$ for all k$]$
  - $\mathbf{P}(s'_{n+1,j}, t) = 1 = \mathbf{P}(s_{n+1,j}, f)$
- $Z = \{\top, \bot, \#\}$ and $obs(s^{\top}_{i,j}) = (s'^{\top}_{i,j}) = \top$, $obs(s^{\bot}_{i,j}) = (s'^{\bot}_{i,j}) = \bot$, $obs(s_{i,j}) = obs(s'_{i,j}) = \# = obs(s_\iota) = obs(t) = obs(f)$.
- $r(t) = 1$ and $r(s) = 0$ for all $s \in S \setminus \{t\}$.

The construction runs in polynomial time. The formal proof of its correctness follows the explanation above precisely.

Approximability Approximating the verification of monitors is also hard, in the sense that it is hard to approximate the maximal risk that a monitor admits.

The maximal risk problem is an optimization problem

Definition 16 (CTP Optimization Problem). *Given an HMM $\mathcal{M}$ with states in S, a unary encoded horizon h, a set of alarm states $A \subseteq S$:*

$$\max_{\tau \in \mathcal{L}(\mathcal{M})} \sum_{\pi \in \Pi^{\mathcal{M}}_{=h} \mid \pi_\downarrow \in A} Pr^{\mathcal{M}}(\pi \mid \tau) \cdot r(\pi_\downarrow)$$

SJ: check notation for paths of length h

SJ: Sync with CTP definition

Lemma 6. *The CTP Optimization Problem is APX-hard.*

This follows from a strict reduction from MAX-3SAT, which is an inapproximable and APX-hard problem [16]. The construction coincides with the reduction in Lemma 5 by observing that the conditional probability to reach a t state is given by $1/m$ times the number of satisfied clauses, i.e., we can compute the maximal number of satisfied clauses in φ on the HMM $\mathcal{M}_\varphi$.

6 Experiments

We empirically evaluate the monitor verification (Sec. 3) and monitor learning (Sec. 4) using our prototype implementation called ToVer. We aim to answer the following questions w.r.t. our verifier and our learner.

Setup. The ToVer tool is implemented in Python and C++ on top of the model checker Storm [17] for data structures and for the MQs in Sec. 4 [20]. We use PAYNT [4] to verify colored MDPs (Def. 7). The learner uses the AAlpy framework [21]. All experiments are run (using exact arithmetic) on a single thread of an AMD Ryzen TRP 5965WX and with a memory limit of 15 GiB.

Luko: Mention we obtain the risk function by model checking some reachability property.

Benchmarks. We take the benchmarks Aiport, Refuel, Evade, and Hidden-Incentive from [20]. We add Icy-Driving, a scaled-up version of the central example and SnL based on the game "Snakes and Ladders". While the existing benchmarks contain many observations, the new benchmarks only have a few different observations. All benchmarks are scalable. Details are given in Appendix A. Code, benchmarks, and logs will be publicly available via the artifact evaluation.

Efficiency of Monitor Verification We investigate scalability along different dimensions and identify the bottlenecks of the method.

Setup. We obtain monitors from the ToVer learning experiments (below, with $\lambda_s = \lambda_u = 0.3$). Every version of the benchmark is run on the first (incorrect) monitor that passed a limited conformance check, an (incorrect) monitor obtained half way the learning process, and the final correct monitor. For each monitor we (1) verify correctness w.r.t. the same λ_s, λ_u and (2) find the minimum in monitor risk and maximum out of monitor risk.

SJ: vague

Results. We present our results in Table 1, which is a subset of a XX benchmarks shown in Appendix B. Generally, we observe that we verify HMMs with hundreds of states and up to hundred thousands of transitions, see benchmarks E-20 to E-23. Benchmarks A-0 to A-3 reflect verification w.r.t. almost trivial monitors, for which it is typically easy to find an counterexample, whereas A-8 to A-11 reflect verification of the same HMM with respect to a larger correct monitor. However, verifying an extreme bound can still be hard, as seen with A-2 compared to A-3. For such benchmarks, missed alarms seem harder to find than false alarms. Increasing the horizon significantly increases the runtime, even for small models, e.g., I-14/15 compared to I-34/35. In all benchmarks, the runtime consists almost exclusively of creating the input to PAYNT (taking the product and creating the MDP) and in running PAYNT. The former runs in polynomial time in the size of the input (see Appendix, whereas the latter uses various heuristics to avoid

SJ: I wonder whether we really want to present case (2) in the main text. It is not solving any of our problem statements, it is typically harder, and it makes the table harder to read. We could still have that in the appendix.

SJ: I would report times $<1s$ as such, and not give such precise timings, unless your timings are that precise

SJ: The column λ_l is not necessary?

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
AirportA-3	A-0	3/10	10	MA	45	113	18	25	450	0.92	0.31	0.41	0.94
AirportA-3	A-1	-	10	MA	45	113	18	25	450	6.24	0.31	5.73	1.00
AirportA-3	A-2	3/10	10	FA	45	113	18	25	450	0.95	0.31	0.43	0.08
AirportA-3	A-3	-	10	FA	45	113	18	25	450	1014.53	0.31	1014.02	0.01
AirportA-3	A-8	3/10	10	MA	45	113	18	128	2304	833.83	0.56	832.93	-
AirportA-3	A-9	-	10	MA	45	113	18	128	2304	788.22	0.56	787.31	0.30
AirportA-3	A-10	3/10	10	FA	45	113	18	128	2304	1143.18	0.53	1142.35	-
AirportA-3	A-11	-	10	FA	45	113	18	128	2304	1106.29	0.55	1105.43	0.30
Evade	E-20	3/10	9	MA	385	1473	325	328	106600	49.11	33.35	8.23	-
Evade	E-21	-	9	MA	385	1473	325	328	106600	50.67	33.39	8.20	0.27
Evade	E-22	3/10	9	FA	385	1473	325	328	106600	49.75	34.26	8.14	-
Evade	E-23	-	9	FA	385	1473	325	328	106600	50.18	33.36	7.88	0.30
Icy-Driving	I-34	3/10	3	FA	3	6	2	8	16	0.02	0.00	0.01	-
Icy-Driving	I-35	-	3	FA	3	6	2	8	16	0.02	0.00	0.01	0.33
Icy-Driving	I-14	3/10	25	FA	3	6	2	2	4	1225.78	0.01	1225.76	-
Icy-Driving	I-15	-	25	FA	3	6	2	2	4	1173.95	0.01	1173.94	0.33

Table 1: Subset of verification results found in Appendix B. The columns give the family name, an ID, learning threshold and horizon, whether we check for missed alarms of ralse alarms. We give the size of the HMM (states, transitions), the number of observations, and the size of the DFA (states, transitions). Furthermore, we list the run time for the complete verification procedure as well as the time spent on transforming the problem into a policy synthesis problem and the policy synthesis in PAYNT. Dashes in the λ_l column indicate finding the extreme instead of verifying a bound. Dashes in λ^{found} indicate no trace was found with a risk above (or below, for FA) the indicated threshold.

the exponential computation time. In the current implementation, except for the (comparably) large Evade benchmarks, the fast majority is spent on PAYNT. The transformation is never the bottleneck.

Efficiency of Monitor Learning We consider the necessity of EQs in the learning, the size of the learned, and the efficiency of learning them, both for $\lambda_s = \lambda_u$ and $\lambda_s \neq \lambda_u$.

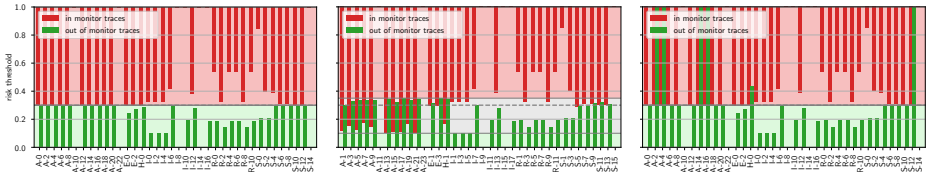
Setup. We use the MAT framework from Sec. 4. Before every EQ, conformance checking (max. 100 samples for λ_s and 100 samples for λ_u, see Sec. 4). As hyper-parameters, we investigate (1) $\lambda_l = 0.3, \lambda_s = 0.35, \lambda_u = 0.1$ and (2) $\lambda_s = \lambda_l = \lambda_u = 0.3$ [7]. We compare with a baseline that does not use EQs, i.e., we use the MAT framework with only conformance checking (max. 100000 samples, cf. Sec. 4) with 100000 samples. Baselines with different numbers of samples are tested in Appendix E.

SJ: What does it mean to run a sample for λ_s?

SJ: The footnote is vague

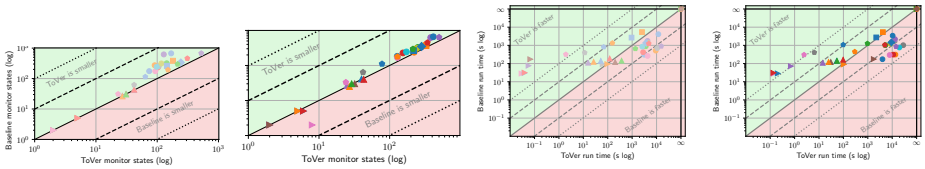
Are the monitors correct? Using a proper EQ, we always learn correct monitors. We validate this experimentally *and* show that the baseline does not always yield correct monitors. Therefore, for any monitor we determine the unsafe trace with

[7] The role of λ_l was evaluated with sampling as can be seen in Appendix E.



(a) ToVer, $\lambda_s = \lambda_l = \lambda_u$ (b) ToVer, $\lambda_s < \lambda_l < \lambda_u$ (c) Baseline

Fig. 10: Actual alarm and actual no-alarm thresholds from monitors learned with ToVer learning and baseline learning. The line between the green and gray area is λ_u, the line between the red and gray area is λ_s. The dotted line is λ_l. Missing bars reflect time-outs.



(a) ToVer with $\lambda_s < \lambda_l < \lambda_u$ (b) ToVer with $\lambda_s = \lambda_l = \lambda_u$ (c) Runtime comparison, $\lambda_s < \lambda_l < \lambda_u$ (d) Runtime comparison, $\lambda_s = \lambda_l = \lambda_u$

Fig. 11: States in learned monitors and runtimes: ToVer vs baseline.

the lowest risk (*actual alarm threshold*, $\lambda_u^{\min}$) and the safe trace with the highest risk (*actual no-alarm threshold*, $\lambda_s^{\max}$). In a correct monitor, we have $\lambda_u^{\min} \geq \lambda_u$ and $\lambda_s^{\max} \leq \lambda_s$. Figures 10a and 10b together and Fig. 10c show $\lambda_u^{\min}$ and $\lambda_s^{\max}$ for ToVer learning and baseline learning respectively. A monitor is correct if its red bar never touches the green area, and the green bar never touches the red area. In 6 out of 38 benchmarks the baseline learns a monitor that misses alarms. No monitors had false alarms.

How big are the monitors? ToVer learns monitors with hundreds of states and tens of thousands transitions, see Figs. 11a and 11b (log-scale!) and Appendix C. For the literature on AAL, these are large automata[31, 2, 26]. Comparing the sizes of the monitors learned using ToVer and the baseline, monitors are smaller (up to 5 times, mostly at least 1.5 times smaller).

SJ: I would simplify the visual of this plot and the other (and increase contrast) by only having black circle/cross for benchmarks.

How fast do we learn the monitors? We compare the runtime of ToVer and baseline in the Figs. 11c and 11d (log-scale!). We remark that only ToVer is guaranteed to be correct. Neither of the two learning algorithms is clearly faster than the other, but ToVer has the potential to significantly accelerate the learning process, despite the high complexity. In particular, it tends to find counterexamples in the early stages much faster. This observation is independent of $\lambda_s = \lambda_u$ or $\lambda_s \neq \lambda_u$. In Appendix ?, we detail where the time is spent. For most benchmarks, the EQ (in particular, PAYNT) is the bottleneck. However,

SJ: How many EQs do we need to learn a monitor?

Luko: Should I add this to the table?

Luko: Add this infromation to a table

for several EVADE benchmarks, most time is spent within L* code. We conjecture this happens as finding counterexamples is simple in these models.

The role of an inconclusive area. We compare between $\lambda_u < \lambda_l < \lambda_s$ and $\lambda_s = \lambda_l = \lambda_u$, i.e., with and without an inconclusive area. The baseline does not actively support such an inconclusive area. With an inconclusive area, more monitors are correct. The learner indeed finds monitors that are 2-5 times smaller (also compare Figures 11a and 11b). For benchmarks ICY-DRIVING, EVADE, and AIRPORT, this also translates to faster runtimes, sometimes by orders of magnitudes.

SJ: check numbers

7 Related

- LTLf
- Predictors or so by Ziemek et al.
- MDPs as Distribution transformers
- Computing conditional probabilities (About reachability conditionals, not ending conditionals)

Theorectical results used in transformation or learning

- Baier *et al.* [11] Conditional probabilities, Badings *et al.* [8] also contains conditioning on evidence, same as Junges, Torfah, and Seshia [20]. However, they all condition on a specific set of evidence, they don't allow the model checker to choose which evidence to condition on.
- Andriushchenko *et al.* [4] PAYNT

Luko: This is a novel idea? Letting the model checker choose which evidence to condition on?

Monitoring on HMMs

Luko: Denk dat dit relevante papers zijn uit de premise paper, maar mijn brein is wat op

- Sistla and Srinivas [24]
- Stoller *et al.* [27]
- Wilcox and Williams [29]
- Sistla, Žefran, and Feng [25]
- Cimatti, Tian, and Tonetta [13]
- [23]
- Woo and Mok [30]

Learn a monitor using AL

- Junges, Seshia, and Torfah [19] Use AL to learn decision trees. Verify them using conformance testing.
- Cairoli, Bortolussi, and Paoletti [12] Use AL to learn a neural network monitor.

Learn a HMM, then do simple reachability to find the probability of error

- Junges, Torfah, and Seshia [20] Premise paper

- Babaee, Gurfinkel, and Fischmeister [7] Learn an HMM and produce a lookup table which for each state calculates the probability to be in a bad state. Now monitoring consists of running the Viterbi state estimation algorith to get a state in the HMM and using the lookup table to check git the risk.
- Babaee, Ganesh, and Sedwards [6] Learn a HMM faster by focussing on rare events?
- Cleaveland *et al.* [14] Model a PA and store risk for each state in PA. Use state estimation at runtime to monitor.

Correctness of monitors

- Aceto *et al.* [1] Defines soundness and completeness for monitors in terms of some property. Thus defines some correctness for monitors. Does not include a gray area. It does include a margin of error.

Questionable relevance

- Sánchez *et al.* [22] Discusses challenges which can be solved using monitoring or RV. Focuses on examples outside typical software applications.
- Zolfagharian *et al.* [32] SMARLA: Machine learned monitor for RL algorithms.
- Henzinger *et al.* [18] Monitor which evaluates fairness not safety.

8 Conclusion and Future Work

SJ: Something about the size of monitor and model together

- Long traces (complexity and practical implementation).
- Unbounded traces (It woudl be great if we can add why this is hard)

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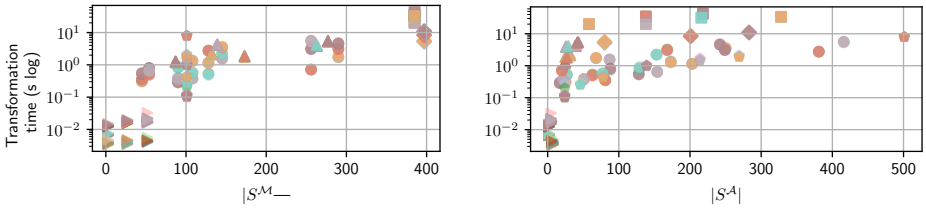


(a) Transformation time steps against HMM states.

(b) Transformation time steps against monitor states.

Fig. 12: Time in the transformation step compared to the size of the HMM $\mathcal{M}$ or the monitor $\mathcal{A}$ for ToVer verification.

A Description of benchmarks

B Results from ToVer Verification Experiments

C Results from ToVer Learning Experiments

Luko: TODO: write about plots

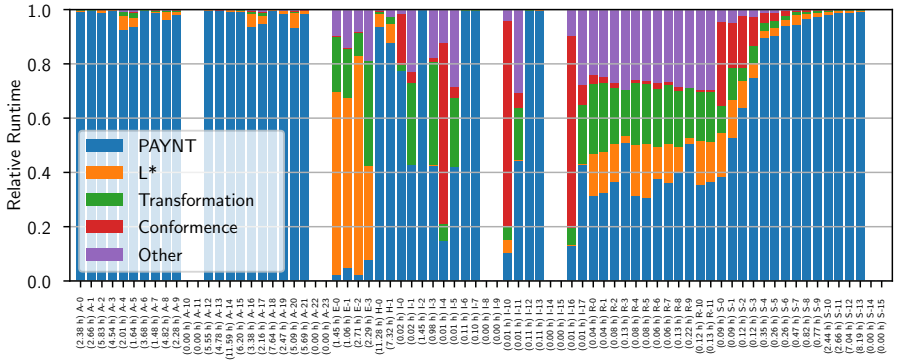


Fig. 13: Runtime division over steps in ToVer learning. Empty bars are HMMs for which learning did not finish.

		Benchmark						ToVer				Baseline			
		λ_u	λ_s	h	$\lvert S\rvert$	$\lvert\mathbf{P}\rvert$	$\lvert Z\rvert$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$
Airport	A-0	3/10	3/10	10	45	113	18	3983.53	128	0.30	0.30	172.76	183	0.30	0.30
Airport	A-1	1/10	7/20	10	45	113	18	3112.38	74	0.12	0.30	304.18	175	0.30	0.30
Airport	A-2	3/10	3/10	10	88	244	32	13530.51	222	0.30	0.30	311.28	238	0.30	1.00

		Benchmark						ToVer				Baseline			
		λ_u	λ_s	h	$\lvert S\rvert$	$\lvert\mathbf{P}\rvert$	$\lvert Z\rvert$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$
Airport	A-3	1/10	7/20	10	88	244	32	9255.42	91	0.15	0.33	513.02	273	0.30	1.00
Airport	A-4	3/10	3/10	10	145	423	50	5222.89	250	0.30	0.30	1029.72	326	0.30	1.00
Airport	A-5	1/10	7/20	10	145	423	50	3612.68	143	0.12	0.34	889.61	317	0.30	1.00
Airport	A-6	3/10	3/10	10	54	150	18	10992.00	154	0.30	0.30	309.27	228	0.30	0.30
Airport	A-7	1/10	7/20	10	54	150	18	2820.88	103	0.18	0.34	421.39	233	0.30	0.30
Airport	A-8	3/10	3/10	10	128	440	32	12964.04	381	0.30	0.30	2167.82	626	0.30	0.30
Airport	A-9	1/10	7/20	10	128	440	32	4275.92	128	0.15	0.33	1767.34	621	0.30	0.30
Airport	A-10	3/10	3/10	10	235	917	50	-	-	-	-	-	-	-	-
Airport	A-11	1/10	7/20	10	235	917	50	-	-	-	-	-	-	-	-
Airport	A-12	3/10	3/10	10	90	334	18	7838.16	129	0.30	0.30	293.90	168	0.30	0.30
Airport	A-13	1/10	7/20	10	90	334	18	4297.92	93	0.11	0.35	261.75	175	0.30	0.30
Airport	A-14	3/10	3/10	10	176	724	32	28299.34	217	0.30	0.30	1020.09	286	0.30	1.00
Airport	A-15	1/10	7/20	10	176	724	32	10621.05	106	0.11	0.32	703.50	255	0.30	1.00
Airport	A-16	3/10	3/10	10	290	1258	50	8544.17	242	0.30	0.30	1257.80	308	0.30	1.00
Airport	A-17	1/10	7/20	10	290	1258	50	4358.92	117	0.11	0.35	1348.91	313	0.30	1.00
Airport	A-18	3/10	3/10	10	108	432	18	19611.01	173	0.30	0.30	762.61	244	0.30	0.30
Airport	A-19	1/10	7/20	10	108	432	18	2602.35	98	0.16	0.33	852.17	243	0.30	0.30
Airport	A-20	3/10	3/10	10	256	1240	32	10403.48	416	0.30	0.30	3402.42	659	0.30	0.30
Airport	A-21	1/10	7/20	10	256	1240	32	11853.95	169	0.10	0.34	4108.96	673	0.30	0.30
Airport	A-22	3/10	3/10	10	470	2550	50	-	-	-	-	-	-	-	-
Airport	A-23	1/10	7/20	10	470	2550	50	-	-	-	-	-	-	-	-
Evade	E-0	3/10	3/10	8	385	1473	325	2214.51	218	0.30	0.24	2732.00	264	0.30	0.24
Evade	E-1	1/10	7/20	8	385	1473	325	944.43	144	0.30	0.31	2665.76	261	0.30	0.24
Evade	E-2	3/10	3/10	9	385	1473	325	4134.45	328	0.30	0.27	5320.22	369	0.30	0.27
Evade	E-3	1/10	7/20	9	385	1473	325	2534.33	181	0.30	0.35	5360.52	383	0.30	0.27
Hidden-Incentive	H-0	3/10	3/10	10	397	1649	100	9190.18	283	0.30	0.28	1448.18	348	0.30	0.43
Hidden-Incentive	H-1	1/10	7/20	10	397	1649	100	3740.06	243	0.17	0.34	1360.26	338	0.30	0.43
Icy-Driving	I-0	3/10	3/10	10	3	6	2	0.70	2	0.33	0.10	71.39	2	0.33	0.10
Icy-Driving	I-1	1/10	7/20	10	3	6	2	0.06	2	0.33	0.10	76.19	2	0.33	0.10
Icy-Driving	I-2	3/10	3/10	25	3	6	2	1750.09	2	0.33	0.10	175.51	2	0.33	0.10
Icy-Driving	I-3	1/10	7/20	25	3	6	2	0.07	2	0.33	0.10	172.98	2	0.33	0.10
Icy-Driving	I-4	3/10	3/10	3	3	6	2	0.14	8	0.33	0.10	27.81	2	0.33	0.10
Icy-Driving	I-5	1/10	7/20	3	3	6	2	0.03	2	0.33	0.10	28.28	2	0.33	0.10
Icy-Driving	I-6	3/10	3/10	10	27	125	2	102.05	5	0.42	0.30	92.19	5	0.42	0.30
Icy-Driving	I-7	1/10	7/20	10	27	125	2	61.50	5	0.42	0.30	91.48	5	0.42	0.30
Icy-Driving	I-8	3/10	3/10	25	27	125	2	-	-	-	-	-	-	-	-
Icy-Driving	I-9	1/10	7/20	25	27	125	2	-	-	-	-	-	-	-	-
Icy-Driving	I-10	3/10	3/10	3	27	125	2	0.20	6	1.00	0.19	28.60	5	1.00	0.19
Icy-Driving	I-11	1/10	7/20	3	27	125	2	0.04	5	1.00	0.19	30.03	5	1.00	0.19
Icy-Driving	I-12	3/10	3/10	10	52	250	2	102.18	5	0.39	0.28	90.23	5	0.39	0.28
Icy-Driving	I-13	1/10	7/20	10	52	250	2	95.72	5	0.39	0.28	93.22	5	0.39	0.28
Icy-Driving	I-14	3/10	3/10	25	52	250	2	-	-	-	-	-	-	-	-
Icy-Driving	I-15	1/10	7/20	25	52	250	2	-	-	-	-	-	-	-	-
Icy-Driving	I-16	3/10	3/10	3	52	250	2	0.14	6	1.00	0.19	30.86	5	1.00	0.19
Icy-Driving	I-17	1/10	7/20	3	52	250	2	0.04	5	1.00	0.19	31.43	5	1.00	0.19
Refuel	R-0	3/10	3/10	10	87	871	63	14.32	25	0.54	0.19	109.19	27	0.54	0.19
Refuel	R-1	1/10	7/20	10	87	871	63	15.68	27	0.54	0.19	110.84	27	0.54	0.19
Refuel	R-2	3/10	3/10	10	132	1798	72	30.38	32	0.32	0.14	121.21	30	0.32	0.14
Refuel	R-3	1/10	7/20	10	132	1798	72	201.44	31	0.32	0.14	120.02	31	0.32	0.14
Refuel	R-4	3/10	3/10	10	139	1628	107	52.75	40	0.54	0.19	139.53	41	0.54	0.19
Refuel	R-5	1/10	7/20	10	139	1628	107	52.02	42	0.54	0.19	140.19	42	0.54	0.19
Refuel	R-6	3/10	3/10	10	173	3429	63	24.68	27	0.54	0.19	121.45	25	0.54	0.19
Refuel	R-7	1/10	7/20	10	173	3429	63	26.42	27	0.54	0.19	123.68	26	0.54	0.19
Refuel	R-8	3/10	3/10	10	263	7127	72	54.80	29	0.32	0.14	139.59	30	0.32	0.14
Refuel	R-9	1/10	7/20	10	263	7127	72	368.99	31	0.32	0.14	143.13	30	0.32	0.14
Refuel	R-10	3/10	3/10	10	277	6415	107	97.30	43	0.54	0.19	162.26	40	0.54	0.19
Refuel	R-11	1/10	7/20	10	277	6415	107	120.52	42	0.54	0.19	188.77	43	0.54	0.19
SnL	S-0	3/10	3/10	10	101	502	4	2.44	24	0.85	0.21	304.14	33	0.85	0.21
SnL	S-1	1/10	7/20	10	101	502	4	1.92	23	0.85	0.21	317.51	31	0.85	0.21
SnL	S-2	3/10	3/10	11	101	502	4	6.43	42	0.40	0.21	404.83	64	0.40	0.21
SnL	S-3	1/10	7/20	11	101	502	4	11.39	42	0.40	0.21	390.89	62	0.40	0.21
SnL	S-4	3/10	3/10	12	101	502	4	97.99	78	0.39	0.29	1075.97	111	0.39	0.29

		Benchmark						ToVer				Baseline			
		λ_u	λ_s	h	$\lvert S\rvert$	$\lvert\mathbf{P}\rvert$	$\lvert Z\rvert$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$	Time (s)	$\lvert\mathcal{A}\rvert$	$\lambda_u^{\min}$	$\lambda_s^{\max}$
SnL	S-5	1/10	7/20	12	101	502	4	68.09	64	0.29	0.29	812.45	114	0.39	0.29
SnL	S-6	3/10	3/10	13	101	502	4	291.77	129	0.31	0.29	448.13	178	0.31	0.29
SnL	S-7	1/10	7/20	13	101	502	4	155.05	150	0.31	0.29	1334.71	193	0.31	0.29
SnL	S-8	3/10	3/10	14	101	502	4	962.41	214	0.31	0.29	1259.73	295	0.31	0.29
SnL	S-9	1/10	7/20	14	101	502	4	1358.30	215	0.31	0.32	656.67	298	0.31	0.29
SnL	S-10	3/10	3/10	15	101	502	4	4995.18	343	0.31	0.29	1047.12	448	0.31	0.29
SnL	S-11	1/10	7/20	15	101	502	4	4797.06	310	0.31	0.32	1240.98	459	0.31	0.29
SnL	S-12	3/10	3/10	16	101	502	4	14325.03	501	0.30	0.29	860.93	638	0.30	1.00
SnL	S-13	1/10	7/20	16	101	502	4	15589.67	523	0.30	0.31	898.79	677	0.30	0.29
SnL	S-14	3/10	3/10	17	101	502	4	-	-	-	-	-	-	-	-
SnL	S-15	1/10	7/20	17	101	502	4	-	-	-	-	-	-	-	-

D Results from Verification Experiment

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert\mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert\mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
AirportA-3	A-0	3/10	10	MA	45	113	18	25	450	0.92	0.31	0.41	0.94
AirportA-3	A-1	-	10	MA	45	113	18	25	450	6.24	0.31	5.73	1.00
AirportA-3	A-2	3/10	10	FA	45	113	18	25	450	0.95	0.31	0.43	0.08
AirportA-3	A-3	-	10	FA	45	113	18	25	450	1014.53	0.31	1014.02	0.01
AirportA-3	A-4	3/10	10	MA	45	113	18	81	1458	85.24	0.35	84.64	1.00
AirportA-3	A-5	-	10	MA	45	113	18	81	1458	80.27	0.35	79.67	1.00
AirportA-3	A-6	3/10	10	FA	45	113	18	81	1458	1.41	0.36	0.81	0.14
AirportA-3	A-7	-	10	FA	45	113	18	81	1458	1310.78	0.35	1310.20	0.01
AirportA-3	A-8	3/10	10	MA	45	113	18	128	2304	833.83	0.56	832.93	-
AirportA-3	A-9	-	10	MA	45	113	18	128	2304	788.22	0.56	787.31	0.30
AirportA-3	A-10	3/10	10	FA	45	113	18	128	2304	1143.18	0.53	1142.35	-
AirportA-3	A-11	-	10	FA	45	113	18	128	2304	1106.29	0.55	1105.43	0.30
AirportA-3	A-12	3/10	10	MA	145	423	50	87	4350	3.71	1.58	1.38	1.00
AirportA-3	A-13	-	10	MA	145	423	50	87	4350	3.79	1.56	1.47	1.00
AirportA-3	A-14	3/10	10	FA	145	423	50	87	4350	11.48	1.56	9.20	0.30
AirportA-3	A-15	-	10	FA	145	423	50	87	4350	707.37	1.55	705.02	0.18
AirportA-3	A-16	3/10	10	MA	145	423	50	153	7650	4.84	2.23	1.74	1.00
AirportA-3	A-17	-	10	MA	145	423	50	153	7650	4.95	2.23	1.88	1.00
AirportA-3	A-18	3/10	10	FA	145	423	50	153	7650	14.34	2.22	11.28	0.30
AirportA-3	A-19	-	10	FA	145	423	50	153	7650	617.78	2.23	614.75	0.03
AirportA-3	A-20	3/10	10	MA	145	423	50	250	12500	219.17	3.52	214.56	-
AirportA-3	A-21	-	10	MA	145	423	50	250	12500	210.95	3.53	206.34	0.30
AirportA-3	A-22	3/10	10	FA	145	423	50	250	12500	487.30	3.51	482.82	-
AirportA-3	A-23	-	10	FA	145	423	50	250	12500	468.35	3.67	463.67	0.30
AirportA-7	A-24	3/10	10	MA	54	150	18	63	1134	18.60	0.53	17.74	0.32
AirportA-7	A-25	-	10	MA	54	150	18	63	1134	334.47	0.52	333.63	0.37
AirportA-7	A-26	3/10	10	FA	54	150	18	63	1134	2.79	0.50	2.01	0.29
AirportA-7	A-27	-	10	FA	54	150	18	63	1134	234.05	0.48	233.27	0.08
AirportA-7	A-28	3/10	10	MA	54	150	18	88	1584	107.41	0.78	106.15	0.33
AirportA-7	A-29	-	10	MA	54	150	18	88	1584	456.47	0.78	455.21	0.87
AirportA-7	A-30	3/10	10	FA	54	150	18	88	1584	4.07	0.81	2.80	0.17
AirportA-7	A-31	-	10	FA	54	150	18	88	1584	329.64	0.74	328.46	0.10
AirportA-7	A-32	3/10	10	MA	54	150	18	154	2772	1187.53	0.64	1186.37	-
AirportA-7	A-33	-	10	MA	54	150	18	154	2772	1103.16	0.66	1101.95	0.30
AirportA-7	A-34	3/10	10	FA	54	150	18	154	2772	443.97	0.64	442.87	-
AirportA-7	A-35	-	10	FA	54	150	18	154	2772	415.53	0.65	414.41	0.30
AirportA-7	A-36	3/10	10	MA	128	440	32	27	864	2.19	0.52	1.31	0.42
AirportA-7	A-37	-	10	MA	128	440	32	27	864	192.49	0.54	191.60	0.99
AirportA-7	A-38	3/10	10	FA	128	440	32	27	864	2.14	0.52	1.27	0.20
AirportA-7	A-39	-	10	FA	128	440	32	27	864	151.51	0.48	150.70	0.00
AirportA-7	A-40	3/10	10	MA	128	440	32	203	6496	12.55	1.14	10.63	0.68
AirportA-7	A-41	-	10	MA	128	440	32	203	6496	411.32	1.17	409.35	0.68

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
AirportA-7	A-42	3/10	10	FA	128	440	32	203	6496	4.84	1.15	2.96	0.13
AirportA-7	A-43	-	10	FA	128	440	32	203	6496	343.71	1.14	341.86	0.07
AirportA-7	A-44	3/10	10	MA	128	440	32	381	12192	624.65	2.73	620.57	-
AirportA-7	A-45	-	10	MA	128	440	32	381	12192	597.28	2.79	593.25	0.30
AirportA-7	A-46	3/10	10	FA	128	440	32	381	12192	1329.33	2.76	1325.47	-
AirportA-7	A-47	-	10	FA	128	440	32	381	12192	1306.39	2.72	1302.57	0.30
AirportB-3	A-48	3/10	10	MA	90	334	18	17	306	1.57	0.30	0.86	0.97
AirportB-3	A-49	-	10	MA	90	334	18	17	306	10.14	0.30	9.47	1.00
AirportB-3	A-50	3/10	10	FA	90	334	18	17	306	1.75	0.28	1.08	0.09
AirportB-3	A-51	-	10	FA	90	334	18	17	306	1501.45	0.30	1500.82	0.09
AirportB-3	A-52	3/10	10	MA	90	334	18	51	918	82.06	0.39	81.23	0.59
AirportB-3	A-53	-	10	MA	90	334	18	51	918	494.71	0.37	493.92	1.00
AirportB-3	A-54	3/10	10	FA	90	334	18	51	918	85.12	0.38	84.34	0.11
AirportB-3	A-55	-	10	FA	90	334	18	51	918	4822.51	0.38	4821.68	0.11
AirportB-3	A-56	3/10	10	MA	90	334	18	129	2322	1378.01	0.86	1376.39	-
AirportB-3	A-57	-	10	MA	90	334	18	129	2322	1361.23	0.86	1359.61	0.30
AirportB-3	A-58	3/10	10	FA	90	334	18	129	2322	3130.67	0.81	3129.17	-
AirportB-3	A-59	-	10	FA	90	334	18	129	2322	3016.85	0.92	3015.22	0.30
AirportB-3	A-60	3/10	10	MA	290	1258	50	68	3400	6.01	1.72	2.98	1.00
AirportB-3	A-61	-	10	MA	290	1258	50	68	3400	6.47	1.73	3.38	1.00
AirportB-3	A-62	3/10	10	FA	290	1258	50	68	3400	22.13	1.79	19.03	0.26
AirportB-3	A-63	-	10	FA	290	1258	50	68	3400	564.11	1.77	561.04	0.01
AirportB-3	A-64	3/10	10	MA	290	1258	50	168	8400	14.84	3.22	9.65	1.00
AirportB-3	A-65	-	10	MA	290	1258	50	168	8400	15.39	3.19	10.24	1.00
AirportB-3	A-66	3/10	10	FA	290	1258	50	168	8400	23.41	3.06	18.48	0.16
AirportB-3	A-67	-	10	FA	290	1258	50	168	8400	505.73	3.05	500.83	0.04
AirportB-3	A-68	3/10	10	MA	290	1258	50	242	12100	530.78	4.52	524.03	-
AirportB-3	A-69	-	10	MA	290	1258	50	242	12100	514.83	4.68	507.83	0.30
AirportB-3	A-70	3/10	10	FA	290	1258	50	242	12100	1266.89	4.60	1260.19	-
AirportB-3	A-71	-	10	FA	290	1258	50	242	12100	1335.06	4.71	1328.15	0.30
AirportB-7	A-72	3/10	10	MA	108	432	18	22	396	3.85	0.39	3.04	0.67
AirportB-7	A-73	-	10	MA	108	432	18	22	396	642.08	0.41	641.25	0.67
AirportB-7	A-74	3/10	10	FA	108	432	18	22	396	3.60	0.37	2.84	0.26
AirportB-7	A-75	-	10	FA	108	432	18	22	396	520.12	0.37	519.38	0.13
AirportB-7	A-76	3/10	10	MA	108	432	18	77	1386	568.81	0.57	567.46	0.32
AirportB-7	A-77	-	10	MA	108	432	18	77	1386	1343.02	0.57	1341.67	0.38
AirportB-7	A-78	3/10	10	FA	108	432	18	77	1386	4.64	0.56	3.34	0.24
AirportB-7	A-79	-	10	FA	108	432	18	77	1386	146.54	0.58	145.23	0.19
AirportB-7	A-80	3/10	10	MA	108	432	18	173	3114	2638.71	1.35	2635.97	-
AirportB-7	A-81	-	10	MA	108	432	18	173	3114	2542.38	1.32	2539.66	0.30
AirportB-7	A-82	3/10	10	FA	108	432	18	173	3114	478.38	1.28	475.86	-
AirportB-7	A-83	-	10	FA	108	432	18	173	3114	450.00	1.28	447.48	0.30
AirportB-7	A-84	3/10	10	MA	256	1240	32	20	640	3.71	0.72	2.34	0.69
AirportB-7	A-85	-	10	MA	256	1240	32	20	640	435.39	0.71	434.02	0.99
AirportB-7	A-86	3/10	10	FA	256	1240	32	20	640	57.80	0.71	56.47	0.23
AirportB-7	A-87	-	10	FA	256	1240	32	20	640	187.32	0.70	186.00	0.11
AirportB-7	A-88	3/10	10	MA	256	1240	32	249	7968	11.85	3.13	6.73	0.31
AirportB-7	A-89	-	10	MA	256	1240	32	249	7968	750.32	3.03	745.26	0.63
AirportB-7	A-90	3/10	10	FA	256	1240	32	249	7968	16.69	3.10	11.70	0.12
AirportB-7	A-91	-	10	FA	256	1240	32	249	7968	1341.91	3.24	1336.81	0.08
AirportB-7	A-92	3/10	10	MA	256	1240	32	416	13312	1142.12	5.55	1133.74	-
AirportB-7	A-93	-	10	MA	256	1240	32	416	13312	1124.68	5.30	1116.56	0.30
AirportB-7	A-94	3/10	10	FA	256	1240	32	416	13312	724.41	5.59	716.21	-
AirportB-7	A-95	-	10	FA	256	1240	32	416	13312	700.22	5.43	692.14	0.30
Evade	E-0	3/10	8	MA	385	1473	325	58	18850	29.20	19.91	4.35	1.00
Evade	E-1	-	8	MA	385	1473	325	58	18850	28.95	19.74	4.28	1.00
Evade	E-2	3/10	8	FA	385	1473	325	58	18850	28.46	20.17	4.37	-
Evade	E-3	-	8	FA	385	1473	325	58	18850	30.40	20.39	4.96	0.30
Evade	E-4	3/10	8	MA	385	1473	325	138	44850	43.84	34.65	3.97	1.00
Evade	E-5	-	8	MA	385	1473	325	138	44850	43.86	34.78	3.97	1.00
Evade	E-6	3/10	8	FA	385	1473	325	138	44850	42.39	34.89	3.97	-
Evade	E-7	-	8	FA	385	1473	325	138	44850	43.01	34.19	4.00	0.30
Evade	E-8	3/10	8	MA	385	1473	325	218	70850	48.76	42.16	3.41	-

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
Evade	E-9	-	8	MA	385	1473	325	218	70850	51.66	43.12	3.86	0.24
Evade	E-10	3/10	8	FA	385	1473	325	218	70850	49.04	42.65	3.37	-
Evade	E-11	-	8	FA	385	1473	325	218	70850	49.57	42.04	3.18	0.30
Evade	E-12	3/10	9	MA	385	1473	325	138	44850	38.70	20.02	9.13	1.00
Evade	E-13	-	9	MA	385	1473	325	138	44850	39.00	20.42	8.84	1.00
Evade	E-14	3/10	9	FA	385	1473	325	138	44850	37.63	20.30	9.05	-
Evade	E-15	-	9	FA	385	1473	325	138	44850	39.46	19.72	10.52	0.30
Evade	E-16	3/10	9	MA	385	1473	325	216	70200	49.36	31.65	8.41	1.00
Evade	E-17	-	9	MA	385	1473	325	216	70200	49.69	31.51	8.49	1.00
Evade	E-18	3/10	9	FA	385	1473	325	216	70200	48.44	32.21	8.46	-
Evade	E-19	-	9	FA	385	1473	325	216	70200	50.80	31.99	9.66	0.30
Evade	E-20	3/10	9	MA	385	1473	325	328	106600	49.11	33.35	8.23	-
Evade	E-21	-	9	MA	385	1473	325	328	106600	50.67	33.39	8.20	0.27
Evade	E-22	3/10	9	FA	385	1473	325	328	106600	49.75	34.26	8.14	-
Evade	E-23	-	9	FA	385	1473	325	328	106600	50.18	33.36	7.88	0.30
Hidden-Incentive	H-0	3/10	10	MA	397	1649	100	80	8000	14.67	5.20	5.52	1.00
Hidden-Incentive	H-1	-	10	MA	397	1649	100	80	8000	14.96	5.37	5.74	1.00
Hidden-Incentive	H-2	3/10	10	FA	397	1649	100	80	8000	16.20	5.63	6.80	0.14
Hidden-Incentive	H-3	-	10	FA	397	1649	100	80	8000	24.65	5.22	15.70	0.14
Hidden-Incentive	H-4	3/10	10	MA	397	1649	100	201	20100	17.67	8.47	5.67	1.00
Hidden-Incentive	H-5	-	10	MA	397	1649	100	201	20100	18.35	8.62	6.26	1.00
Hidden-Incentive	H-6	3/10	10	FA	397	1649	100	201	20100	21.21	8.31	9.60	0.22
Hidden-Incentive	H-7	-	10	FA	397	1649	100	201	20100	3123.98	8.40	3112.28	0.22
Hidden-Incentive	H-8	3/10	10	MA	397	1649	100	283	28300	1274.37	10.89	1260.42	-
Hidden-Incentive	H-9	-	10	MA	397	1649	100	283	28300	1290.20	10.90	1276.26	0.28
Hidden-Incentive	H-10	3/10	10	FA	397	1649	100	283	28300	6686.27	11.00	6672.29	-
Hidden-Incentive	H-11	-	10	FA	397	1649	100	283	28300	8822.49	10.86	8808.66	0.30
Icy-Driving	I-0	3/10	10	MA	3	6	2	2	4	0.02	0.01	0.01	-
Icy-Driving	I-1	-	10	MA	3	6	2	2	4	0.02	0.01	0.01	0.10
Icy-Driving	I-2	3/10	10	FA	3	6	2	2	4	0.55	0.01	0.54	-
Icy-Driving	I-3	-	10	FA	3	6	2	2	4	0.53	0.01	0.51	0.33
Icy-Driving	I-4	3/10	10	MA	3	6	2	2	4	0.02	0.01	0.01	-
Icy-Driving	I-5	-	10	MA	3	6	2	2	4	0.02	0.01	0.01	0.10
Icy-Driving	I-6	3/10	10	FA	3	6	2	2	4	0.56	0.01	0.55	-
Icy-Driving	I-7	-	10	FA	3	6	2	2	4	0.53	0.01	0.51	0.33
Icy-Driving	I-8	3/10	10	MA	3	6	2	2	4	0.02	0.01	0.01	-
Icy-Driving	I-9	-	10	MA	3	6	2	2	4	0.02	0.01	0.01	0.10
Icy-Driving	I-10	3/10	10	FA	3	6	2	2	4	0.55	0.01	0.54	-
Icy-Driving	I-11	-	10	FA	3	6	2	2	4	0.53	0.01	0.51	0.33
Icy-Driving	I-12	3/10	25	MA	3	6	2	2	4	0.03	0.01	0.01	-
Icy-Driving	I-13	-	25	MA	3	6	2	2	4	0.03	0.01	0.01	0.10
Icy-Driving	I-14	3/10	25	FA	3	6	2	2	4	1225.78	0.01	1225.76	-
Icy-Driving	I-15	-	25	FA	3	6	2	2	4	1173.95	0.01	1173.94	0.33
Icy-Driving	I-16	3/10	25	MA	3	6	2	2	4	0.03	0.01	0.01	-
Icy-Driving	I-17	-	25	MA	3	6	2	2	4	0.03	0.01	0.01	0.10
Icy-Driving	I-18	3/10	25	FA	3	6	2	2	4	1201.62	0.01	1201.60	-
Icy-Driving	I-19	-	25	FA	3	6	2	2	4	1215.28	0.01	1215.26	0.33
Icy-Driving	I-20	3/10	25	MA	3	6	2	2	4	0.03	0.01	0.01	-
Icy-Driving	I-21	-	25	MA	3	6	2	2	4	0.03	0.01	0.01	0.10
Icy-Driving	I-22	3/10	25	FA	3	6	2	2	4	1262.11	0.01	1262.10	-
Icy-Driving	I-23	-	25	FA	3	6	2	2	4	1204.32	0.01	1204.30	0.33
Icy-Driving	I-24	3/10	3	MA	3	6	2	8	16	0.01	0.00	0.01	-
Icy-Driving	I-25	-	3	MA	3	6	2	8	16	0.02	0.00	0.01	0.10
Icy-Driving	I-26	3/10	3	FA	3	6	2	8	16	0.02	0.00	0.01	-
Icy-Driving	I-27	-	3	FA	3	6	2	8	16	0.02	0.00	0.01	0.33
Icy-Driving	I-28	3/10	3	MA	3	6	2	8	16	0.01	0.00	0.01	-
Icy-Driving	I-29	-	3	MA	3	6	2	8	16	0.02	0.00	0.01	0.10
Icy-Driving	I-30	3/10	3	FA	3	6	2	8	16	0.02	0.00	0.01	-
Icy-Driving	I-31	-	3	FA	3	6	2	8	16	0.02	0.00	0.01	0.33
Icy-Driving	I-32	3/10	3	MA	3	6	2	8	16	0.01	0.00	0.01	-
Icy-Driving	I-33	-	3	MA	3	6	2	8	16	0.02	0.00	0.01	0.10
Icy-Driving	I-34	3/10	3	FA	3	6	2	8	16	0.02	0.00	0.01	-
Icy-Driving	I-35	-	3	FA	3	6	2	8	16	0.02	0.00	0.01	0.33

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
Icy-Driving	I-36	3/10	10	MA	27	125	2	5	10	0.75	0.02	0.73	-
Icy-Driving	I-37	-	10	MA	27	125	2	5	10	0.73	0.02	0.71	0.30
Icy-Driving	I-38	3/10	10	FA	27	125	2	5	10	99.46	0.02	99.44	-
Icy-Driving	I-39	-	10	FA	27	125	2	5	10	93.26	0.02	93.23	0.42
Icy-Driving	I-40	3/10	10	MA	27	125	2	5	10	0.73	0.02	0.71	-
Icy-Driving	I-41	-	10	MA	27	125	2	5	10	0.72	0.02	0.70	0.30
Icy-Driving	I-42	3/10	10	FA	27	125	2	5	10	98.13	0.02	98.11	-
Icy-Driving	I-43	-	10	FA	27	125	2	5	10	94.41	0.02	94.39	0.42
Icy-Driving	I-44	3/10	10	MA	27	125	2	5	10	0.73	0.02	0.71	-
Icy-Driving	I-45	-	10	MA	27	125	2	5	10	0.71	0.02	0.68	0.30
Icy-Driving	I-46	3/10	10	FA	27	125	2	5	10	97.12	0.02	97.10	-
Icy-Driving	I-47	-	10	FA	27	125	2	5	10	92.13	0.02	92.10	0.42
Icy-Driving	I-48	3/10	3	MA	27	125	2	6	12	0.02	0.00	0.01	-
Icy-Driving	I-49	-	3	MA	27	125	2	6	12	0.02	0.00	0.01	0.19
Icy-Driving	I-50	3/10	3	FA	27	125	2	6	12	0.01	0.00	0.01	-
Icy-Driving	I-51	-	3	FA	27	125	2	6	12	-	-	-	-
Icy-Driving	I-52	3/10	3	MA	27	125	2	6	12	0.02	0.00	0.01	-
Icy-Driving	I-53	-	3	MA	27	125	2	6	12	0.02	0.00	0.01	0.19
Icy-Driving	I-54	3/10	3	FA	27	125	2	6	12	0.01	0.00	0.01	-
Icy-Driving	I-55	-	3	FA	27	125	2	6	12	-	-	-	-
Icy-Driving	I-56	3/10	3	MA	27	125	2	6	12	0.02	0.00	0.02	-
Icy-Driving	I-57	-	3	MA	27	125	2	6	12	0.03	0.00	0.01	0.19
Icy-Driving	I-58	3/10	3	FA	27	125	2	6	12	0.01	0.00	0.00	-
Icy-Driving	I-59	-	3	FA	27	125	2	6	12	-	-	-	-
Icy-Driving	I-60	3/10	10	MA	52	250	2	5	10	0.81	0.02	0.79	-
Icy-Driving	I-61	-	10	MA	52	250	2	5	10	0.75	0.02	0.73	0.28
Icy-Driving	I-62	3/10	10	FA	52	250	2	5	10	102.11	0.02	102.09	-
Icy-Driving	I-63	-	10	FA	52	250	2	5	10	95.05	0.03	94.96	0.39
Icy-Driving	I-64	3/10	10	MA	52	250	2	5	10	0.72	0.02	0.70	-
Icy-Driving	I-65	-	10	MA	52	250	2	5	10	0.71	0.02	0.69	0.28
Icy-Driving	I-66	3/10	10	FA	52	250	2	5	10	102.30	0.02	102.27	-
Icy-Driving	I-67	-	10	FA	52	250	2	5	10	94.48	0.02	94.45	0.39
Icy-Driving	I-68	3/10	10	MA	52	250	2	5	10	0.75	0.02	0.72	-
Icy-Driving	I-69	-	10	MA	52	250	2	5	10	0.74	0.02	0.71	0.28
Icy-Driving	I-70	3/10	10	FA	52	250	2	5	10	102.25	0.02	102.23	-
Icy-Driving	I-71	-	10	FA	52	250	2	5	10	93.56	0.02	93.53	0.39
Icy-Driving	I-72	3/10	3	MA	52	250	2	6	12	0.02	0.00	0.01	-
Icy-Driving	I-73	-	3	MA	52	250	2	6	12	0.03	0.00	0.01	0.19
Icy-Driving	I-74	3/10	3	FA	52	250	2	6	12	0.01	0.00	0.01	-
Icy-Driving	I-75	-	3	FA	52	250	2	6	12	-	-	-	-
Icy-Driving	I-76	3/10	3	MA	52	250	2	6	12	0.02	0.00	0.01	-
Icy-Driving	I-77	-	3	MA	52	250	2	6	12	0.03	0.00	0.01	0.19
Icy-Driving	I-78	3/10	3	FA	52	250	2	6	12	0.01	0.00	0.01	-
Icy-Driving	I-79	-	3	FA	52	250	2	6	12	-	-	-	-
Icy-Driving	I-80	3/10	3	MA	52	250	2	6	12	0.02	0.00	0.01	-
Icy-Driving	I-81	-	3	MA	52	250	2	6	12	0.03	0.01	0.01	0.19
Icy-Driving	I-82	3/10	3	FA	52	250	2	6	12	0.01	0.00	0.01	-
Icy-Driving	I-83	-	3	FA	52	250	2	6	12	-	-	-	-
Refuel	R-0	3/10	10	MA	87	871	63	24	1512	4.00	1.22	1.72	1.00
Refuel	R-1	-	10	MA	87	871	63	24	1512	4.22	1.21	1.90	1.00
Refuel	R-2	3/10	10	FA	87	871	63	24	1512	3.80	1.20	1.49	-
Refuel	R-3	-	10	FA	87	871	63	24	1512	12.02	1.26	9.65	0.54
Refuel	R-4	3/10	10	MA	87	871	63	24	1512	3.98	1.18	1.72	1.00
Refuel	R-5	-	10	MA	87	871	63	24	1512	4.18	1.23	1.86	1.00
Refuel	R-6	3/10	10	FA	87	871	63	24	1512	3.73	1.23	1.44	-
Refuel	R-7	-	10	FA	87	871	63	24	1512	11.89	1.27	9.50	0.54
Refuel	R-8	3/10	10	MA	87	871	63	25	1575	3.62	1.24	1.34	-
Refuel	R-9	-	10	MA	87	871	63	25	1575	3.60	1.21	1.33	0.19
Refuel	R-10	3/10	10	FA	87	871	63	25	1575	3.91	1.31	1.50	-
Refuel	R-11	-	10	FA	87	871	63	25	1575	11.91	1.22	9.55	0.54
Refuel	R-12	3/10	10	MA	132	1798	72	31	2232	9.37	2.30	4.41	1.00
Refuel	R-13	-	10	MA	132	1798	72	31	2232	9.66	2.08	4.84	1.00
Refuel	R-14	3/10	10	FA	132	1798	72	31	2232	8.27	2.16	3.46	-
Refuel	R-15	-	10	FA	132	1798	72	31	2232	55.78	2.11	50.99	0.32

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
Refuel	R-16	3/10	10	MA	132	1798	72	31	2232	9.38	2.11	4.47	1.00
Refuel	R-17	-	10	MA	132	1798	72	31	2232	9.63	2.08	4.82	1.00
Refuel	R-18	3/10	10	FA	132	1798	72	31	2232	8.18	2.10	3.47	-
Refuel	R-19	-	10	FA	132	1798	72	31	2232	56.10	2.12	51.36	0.32
Refuel	R-20	3/10	10	MA	132	1798	72	32	2304	8.10	2.15	3.33	-
Refuel	R-21	-	10	MA	132	1798	72	32	2304	8.27	2.19	3.37	0.14
Refuel	R-22	3/10	10	FA	132	1798	72	32	2304	8.33	2.17	3.43	-
Refuel	R-23	-	10	FA	132	1798	72	32	2304	55.13	2.16	50.29	0.32
Refuel	R-24	3/10	10	MA	139	1628	107	39	4173	14.28	4.07	5.90	1.00
Refuel	R-25	-	10	MA	139	1628	107	39	4173	14.65	4.28	6.06	1.00
Refuel	R-26	3/10	10	FA	139	1628	107	39	4173	13.53	4.13	5.16	-
Refuel	R-27	-	10	FA	139	1628	107	39	4173	28.43	4.09	20.02	0.54
Refuel	R-28	3/10	10	MA	139	1628	107	39	4173	14.24	4.10	5.86	1.00
Refuel	R-29	-	10	MA	139	1628	107	39	4173	14.74	4.28	6.11	1.00
Refuel	R-30	3/10	10	FA	139	1628	107	39	4173	13.49	4.09	5.20	-
Refuel	R-31	-	10	FA	139	1628	107	39	4173	28.89	4.12	20.31	0.54
Refuel	R-32	3/10	10	MA	139	1628	107	40	4280	13.71	4.45	5.03	-
Refuel	R-33	-	10	MA	139	1628	107	40	4280	13.28	4.15	4.97	0.19
Refuel	R-34	3/10	10	FA	139	1628	107	40	4280	13.93	4.13	5.25	-
Refuel	R-35	-	10	FA	139	1628	107	40	4280	28.78	4.15	20.37	0.54
RefuelB	R-36	3/10	10	MA	173	3429	63	25	1575	7.48	1.83	3.50	1.00
RefuelB	R-37	-	10	MA	173	3429	63	25	1575	8.16	1.90	4.08	1.00
RefuelB	R-38	3/10	10	FA	173	3429	63	25	1575	7.06	1.88	3.00	-
RefuelB	R-39	-	10	FA	173	3429	63	25	1575	26.62	1.88	22.52	0.54
RefuelB	R-40	3/10	10	MA	173	3429	63	25	1575	7.52	1.85	3.50	1.00
RefuelB	R-41	-	10	MA	173	3429	63	25	1575	8.00	1.88	3.86	1.00
RefuelB	R-42	3/10	10	FA	173	3429	63	25	1575	7.09	1.83	3.04	-
RefuelB	R-43	-	10	FA	173	3429	63	25	1575	26.70	1.90	22.57	0.54
RefuelB	R-44	3/10	10	MA	173	3429	63	27	1701	6.79	1.83	2.79	-
RefuelB	R-45	-	10	MA	173	3429	63	27	1701	6.68	1.79	2.75	0.19
RefuelB	R-46	3/10	10	FA	173	3429	63	27	1701	6.96	1.78	3.02	-
RefuelB	R-47	-	10	FA	173	3429	63	27	1701	26.94	1.86	22.79	0.54
RefuelB	R-48	3/10	10	MA	263	7127	72	26	1872	17.77	3.88	8.72	1.00
RefuelB	R-49	-	10	MA	263	7127	72	26	1872	18.61	3.88	9.56	1.00
RefuelB	R-50	3/10	10	FA	263	7127	72	26	1872	15.95	3.97	6.81	-
RefuelB	R-51	-	10	FA	263	7127	72	26	1872	117.86	3.82	108.91	0.32
RefuelB	R-52	3/10	10	MA	263	7127	72	26	1872	18.11	4.08	8.74	1.00
RefuelB	R-53	-	10	MA	263	7127	72	26	1872	18.79	4.02	9.61	1.00
RefuelB	R-54	3/10	10	FA	263	7127	72	26	1872	15.83	3.82	6.70	-
RefuelB	R-55	-	10	FA	263	7127	72	26	1872	118.27	3.85	109.17	0.32
RefuelB	R-56	3/10	10	MA	263	7127	72	29	2088	16.08	4.19	6.74	-
RefuelB	R-57	-	10	MA	263	7127	72	29	2088	15.88	4.01	6.68	0.14
RefuelB	R-58	3/10	10	FA	263	7127	72	29	2088	16.11	3.99	6.87	-
RefuelB	R-59	-	10	FA	263	7127	72	29	2088	119.01	4.17	109.40	0.32
RefuelB	R-60	3/10	10	MA	277	6415	107	42	4494	26.23	5.43	12.13	1.00
RefuelB	R-61	-	10	MA	277	6415	107	42	4494	26.50	5.25	12.61	1.00
RefuelB	R-62	3/10	10	FA	277	6415	107	42	4494	24.31	5.45	10.34	-
RefuelB	R-63	-	10	FA	277	6415	107	42	4494	56.28	5.42	42.12	0.54
RefuelB	R-64	3/10	10	MA	277	6415	107	42	4494	25.77	5.39	11.96	1.00
RefuelB	R-65	-	10	MA	277	6415	107	42	4494	26.47	5.23	12.61	1.00
RefuelB	R-66	3/10	10	FA	277	6415	107	42	4494	24.42	5.17	10.65	-
RefuelB	R-67	-	10	FA	277	6415	107	42	4494	56.09	5.23	42.14	0.54
RefuelB	R-68	3/10	10	MA	277	6415	107	43	4601	24.03	5.42	10.07	-
RefuelB	R-69	-	10	MA	277	6415	107	43	4601	23.69	5.36	9.98	0.19
RefuelB	R-70	3/10	10	FA	277	6415	107	43	4601	24.83	5.49	10.60	-
RefuelB	R-71	-	10	FA	277	6415	107	43	4601	55.74	5.40	41.56	0.54
SnL	S-0	3/10	10	MA	101	502	4	24	96	0.63	0.11	0.48	-
SnL	S-1	-	10	MA	101	502	4	24	96	2.67	0.28	2.34	0.21
SnL	S-2	3/10	10	FA	101	502	4	24	96	0.55	0.11	0.41	-
SnL	S-3	-	10	FA	101	502	4	24	96	0.57	0.12	0.40	0.85
SnL	S-4	3/10	10	MA	101	502	4	24	96	0.72	0.20	0.48	-
SnL	S-5	-	10	MA	101	502	4	24	96	2.48	0.12	2.31	0.21
SnL	S-6	3/10	10	FA	101	502	4	24	96	0.57	0.12	0.41	-

		Benchmark								ToVer			
		λ_l	h	MA/FA	$\lvert S^{\mathcal{M}}\rvert$	$\lvert \mathbf{P}^{\mathcal{M}}\rvert$	$\lvert Z\rvert$	$\lvert S^{\mathcal{A}}\rvert$	$\lvert \mathbf{P}^{\mathcal{A}}\rvert$	Time (s)	Trans (s)	PAYNT (s)	λ^{found}
SnL	S-7	-	10	FA	101	502	4	24	96	0.56	0.10	0.41	0.85
SnL	S-8	3/10	10	MA	101	502	4	24	96	0.64	0.11	0.49	-
SnL	S-9	-	10	MA	101	502	4	24	96	2.50	0.12	2.32	0.21
SnL	S-10	3/10	10	FA	101	502	4	24	96	0.55	0.10	0.42	-
SnL	S-11	-	10	FA	101	502	4	24	96	0.55	0.10	0.41	0.85
SnL	S-12	3/10	12	MA	101	502	4	1	4	-	-	-	-
SnL	S-13	-	12	MA	101	502	4	1	4	-	-	-	-
SnL	S-14	3/10	12	FA	101	502	4	1	4	-	-	-	-
SnL	S-15	-	12	FA	101	502	4	1	4	-	-	-	-
SnL	S-16	3/10	12	MA	101	502	4	46	184	17.99	0.25	17.64	-
SnL	S-17	-	12	MA	101	502	4	46	184	26.01	0.29	25.62	0.29
SnL	S-18	3/10	12	FA	101	502	4	46	184	1.75	0.26	1.41	0.00
SnL	S-19	-	12	FA	101	502	4	46	184	1.75	0.26	1.40	0.00
SnL	S-20	3/10	12	MA	101	502	4	78	312	22.58	0.44	22.02	-
SnL	S-21	-	12	MA	101	502	4	78	312	31.09	0.45	30.50	0.29
SnL	S-22	3/10	12	FA	101	502	4	78	312	3.73	0.42	3.21	-
SnL	S-23	-	12	FA	101	502	4	78	312	3.71	0.43	3.14	0.39
SnL	S-24	3/10	14	MA	101	502	4	1	4	-	-	-	-
SnL	S-25	-	14	MA	101	502	4	1	4	-	-	-	-
SnL	S-26	3/10	14	FA	101	502	4	1	4	-	-	-	-
SnL	S-27	-	14	FA	101	502	4	1	4	-	-	-	-
SnL	S-28	3/10	14	MA	101	502	4	139	556	12.32	1.04	10.88	0.82
SnL	S-29	-	14	MA	101	502	4	139	556	175.94	1.02	174.58	1.00
SnL	S-30	3/10	14	FA	101	502	4	139	556	37.99	0.96	36.70	0.29
SnL	S-31	-	14	FA	101	502	4	139	556	112.18	1.01	110.84	0.14
SnL	S-32	3/10	14	MA	101	502	4	214	856	152.92	1.69	150.95	-
SnL	S-33	-	14	MA	101	502	4	214	856	150.38	1.52	148.57	0.29
SnL	S-34	3/10	14	FA	101	502	4	214	856	183.23	1.47	181.55	-
SnL	S-35	-	14	FA	101	502	4	214	856	246.79	1.55	244.99	0.31
SnL	S-36	3/10	16	MA	101	502	4	1	4	-	-	-	-
SnL	S-37	-	16	MA	101	502	4	1	4	-	-	-	-
SnL	S-38	3/10	16	FA	101	502	4	1	4	-	-	-	-
SnL	S-39	-	16	FA	101	502	4	1	4	-	-	-	-
SnL	S-40	3/10	16	MA	101	502	4	269	1076	24.87	1.99	22.26	0.31
SnL	S-41	-	16	MA	101	502	4	269	1076	384.38	2.14	381.55	1.00
SnL	S-42	3/10	16	FA	101	502	4	269	1076	9.05	1.85	6.68	0.00
SnL	S-43	-	16	FA	101	502	4	269	1076	9.22	1.88	6.82	0.00
SnL	S-44	3/10	16	MA	101	502	4	501	2004	1357.81	7.91	1349.03	-
SnL	S-45	-	16	MA	101	502	4	501	2004	1235.13	8.16	1226.04	0.29
SnL	S-46	3/10	16	FA	101	502	4	501	2004	2723.58	7.94	2714.98	-
SnL	S-47	-	16	FA	101	502	4	501	2004	2536.77	7.72	2528.44	0.30

E Results from Baseline Sampling Count Experiment

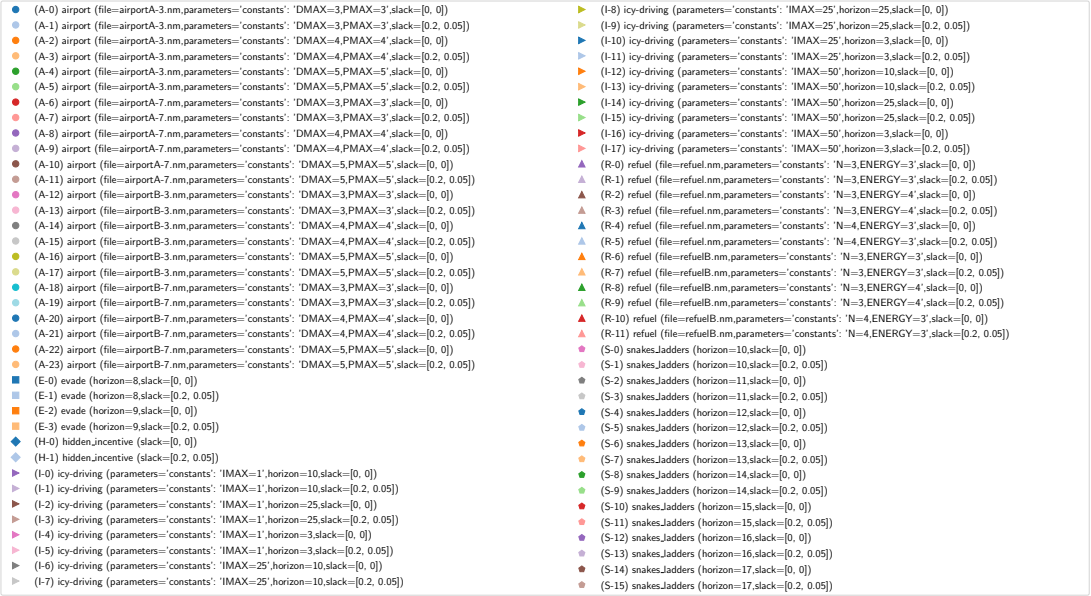
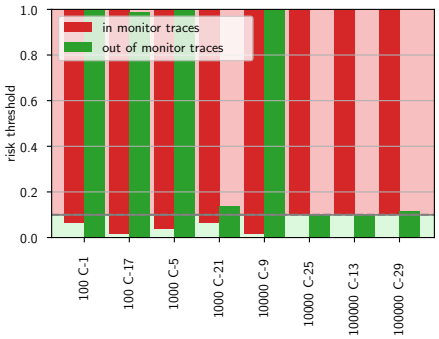


Fig. 14: Legend of symbols used in plots



(a) Risk threshold of 0.1

(b) Risk threshold of 0.2

(c) Risk threshold of 0.4

Fig. 15: Found risk thresholds when using baseline leaning with a different amount of samples in each EQ step. This experiment is done for three different risk thresholds.